[Document(metadata={'producer': 'pdfTeX-1.40.25', 'creator': 'LaTeX with hyperref', 'creationdate': '2024-04-10T21:11:43+00:00', 'source': '..\\data\\pdf\\attention.pdf', 'file_path': '..\\data\\pdf\\attention.pdf', 'total_pages': 15, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2024-04-10T21:11:43+00:00', 'trapped': '', 'modDate': 'D:20240410211143Z', 'creationDate': 'D:20240410211143Z', 'page': 0}, page_content='Provided proper attribution is provided, Google hereby grants permission to\nreproduce the tables and figures in this paper solely for use in journalistic or\nscholarly works.\nAttention Is All You Need\nAshish Vaswani∗ \nGoogle Brain\navaswani@google.com\nNoam Shazeer∗\nGoogle Brain\nnoam@google.com \nNiki Parmar∗\nGoogle Research\nnikip@google.com\nJakob Uszkoreit∗\nGoogle Research \nusz@google.com\nLlion Jones∗\nGoogle Research\nllion@google.com\nAidan N. Gomez∗ † \nUniversity of Toronto\naidan@cs.toronto.edu\nŁukasz Kaiser∗\nGoogle Brain \nlukaszkaiser@google.com\nIllia Polosukhin∗ ‡\nillia.polosukhin@gmail.com\nAbstract\nThe dominant sequence transduction models are based on complex recurrent or\nconvolutional neural networks that include an encoder and a decoder. The best\nperforming models also connect the encoder and decoder through an attention\nmechanism. We propose a new simple network architecture, the Transformer,\nbased solely on attention mechanisms, dispensing with recurrence and convolutions\nentirely. Experiments on two machine translation tasks show these models to\nbe superior in quality while being more parallelizable and requiring significantly \nless time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-\nto-German translation task, improving over the existing best results, including\nensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task,\nour model establishes a new single-model state-of-the-art BLEU score of 41.8 after\ntraining for 3.5 days on eight GPUs, a small fraction of the training costs of the\nbest models from the literature. We show that the Transformer generalizes well to\nother tasks by applying it successfully to English constituency parsing both with\nlarge and limited training data.\n∗Equacontribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started\nthe effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and\nhas been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head \nattention and the parameter-free position representation and became the other person involved in nearly every\ndetail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and\ntensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and\nefficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and \nimplementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating\nour research.\n†Work performed while at Google Brain.\n‡Work performed while at Google Research.\n31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.\narXiv:1706.03762v7 [cs.CL] 2 Aug 2023'),

Document(metadata={'producer': 'pdfTeX-1.40.25', 'creator': 'LaTeX with hyperref', 'creationdate': '2024-04-10T21:11:43+00:00', 'source': '..\\data\\pdf\\attention.pdf', 'file_path': '..\\data\\pdf\\attention.pdf', 'total_pages': 15, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2024-04-10T21:11:43+00:00', 'trapped': '', 'modDate': 'D:20240410211143Z', 'creationDate': 'D:20240410211143Z', 'page': 1}, page_content='1\nIntroduction\nRecurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks\nin particular, have been firmly established as state of the art approaches in sequence modeling and \ntransduction problems such as language modeling and machine translation [35, 2, 5]. Numerous\nefforts have since continued to push the boundaries of recurrent language models and encoder-decoder\narchitectures [38, 24, 15].\nRecurrent models typically factor computation along the symbol positions of the input and output\nsequences. Aligning the positions to steps in computation time, they generate a sequence of hidden\nstates ht, as a function of the previous hidden state ht−1 and the input for position t. This inherently \nsequential nature precludes parallelization within training examples, which becomes critical at longer\nsequence lengths, as memory constraints limit batching across examples. Recent work has achieved\nsignificant improvements in computational efficiency through factorization tricks [21] and conditional\ncomputation [32], while also improving model performance in case of the

latter. The fundamental\nconstraint of sequential computation, however, remains.\nAttention mechanisms have become an integral part of compelling sequence modeling and transduc-\ntion models in various tasks, allowing modeling of dependencies without regard to their distance in\nthe input or output sequences [2, 19]. In all but a few cases [27], however, such attention mechanisms\nare used in conjunction with a recurrent network.\nIn this work we propose the Transformer, a model architecture eschewing recurrence and instead\nrelying entirely on an attention mechanism to draw global dependencies between input and output. \nThe Transformer allows for significantly more parallelization and can reach a new state of the art in\ntranslation quality after being trained for as little as twelve hours on eight P100 GPUs. \n2\nBackground\nThe goal of reducing sequential computation also forms the foundation of the Extended Neural GPU\n[16], ByteNet [18] and ConvS2S [9], all of which use convolutional neural networks as basic building\nblock, computing hidden representations in parallel for all input and output positions. In these models,\nthe number of operations required to relate signals from two arbitrary input or output positions grows\nin the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes\nit more difficult to learn dependencies between distant positions [12]. In the Transformer this is\nreduced to a constant number of operations, albeit at the cost of reduced effective resolution due\nto averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as\ndescribed in section 3.2.\nSelf-attention, sometimes called intra-attention is an attention mechanism relating different positions\nof a single sequence in order to compute a representation of the sequence. Self-attention has been\nused successfully in a variety of tasks including reading comprehension, abstractive summarization,\ntextual entailment and learning task-independent sentence representations [4, 27, 28, 22].\nEnd-to-end memory networks are based on a recurrent attention mechanism instead of sequence-\naligned recurrence and have been shown to perform well on simple-language question answering and\nlanguage modeling tasks [34].\nTo the best of our knowledge, however, the Transformer is the first transduction model relying\nentirely on self-attention to compute representations of its input and output without using sequence-\naligned RNNs or convolution. In the following sections, we will describe the Transformer, motivate\nself-attention and discuss its advantages over models such as [17, 18] and [9].\n3\nModel Architecture\nMost competitive neural sequence transduction models have an encoder-decoder structure [5, 2, 35].\nHere, the encoder maps an input sequence of symbol representations (x1, ..., xn) to a sequence\nof continuous representations z = (z1, ..., zn). Given z, the decoder then generates an output\nsequence (y1, ..., ym) of symbols one element at a time. At each step the model is auto-regressive\n[10], consuming the previously generated symbols as additional input when generating the next.\n2'),

Document(metadata={'producer': 'pdfTeX-1.40.25', 'creator': 'LaTeX with hyperref', 'creationdate': '2024-04-10T21:11:43+00:00', 'source': '..\\data\\pdf\\attention.pdf', 'file_path': '..\\data\\pdf\\attention.pdf', 'total_pages': 15, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2024-04-10T21:11:43+00:00', 'trapped': '', 'modDate': 'D:20240410211143Z', 'creationDate': 'D:20240410211143Z', 'page': 2}, page_content='Figure 1: The Transformer - model architecture.\nThe Transformer follows this overall architecture using stacked self-attention and point-wise, fully\nconnected layers for both the encoder and decoder, shown in the left and right halves of Figure 1,\nrespectively.\n3.1\nEncoder and Decoder Stacks\nEncoder:\nThe encoder is composed of a stack of N = 6 identical layers. Each layer has two\nsub-layers. The first is a multi-head self-attention mechanism, and the second is a simple, position-\nwise fully connected feed-forward network. We employ a residual connection [11] around each of\nthe two sub-layers, followed by layer normalization [1]. That is, the output of each sub-layer is\nLayerNorm(x + Sublayer(x)), where Sublayer(x) is the function implemented by the sub-layer\nitself. To facilitate these residual connections, all sub-layers in the model, as well as the embedding\nlayers, produce outputs of dimension dmodel = 512.\nDecoder:\nThe decoder is also composed of a stack of N = 6 identical layers. In addition to the two\nsub-layers in each encoder layer, the decoder inserts a third sub-layer, which performs multi-head\nattention over the output of the encoder stack. Similar to the encoder, we employ residual connections\naround each of the sub-layers, followed by layer normalization. We also modify the self-attention\nsub-layer in the decoder stack to prevent positions from attending to subsequent positions. This\nmasking,

combined with fact that the output embeddings are offset by one position, ensures that the \npredictions for position i can depend only on the known outputs at positions less than i. \n3.2\nAttention\nAn attention function can be described as mapping a query and a set of key-value pairs to an output,\nwhere the query, keys, values, and output are all vectors. The output is computed as a weighted sum\n3'),

Document(metadata={'producer': 'pdfTeX-1.40.25', 'creator': 'LaTeX with hyperref', 'creationdate': '2024-04-10T21:11:43+00:00', 'source': '..\\data\\pdf\\attention.pdf', 'file_path': '..\\data\\pdf\\attention.pdf', 'total_pages': 15, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2024-04-10T21:11:43+00:00', 'trapped': '', 'modDate': 'D:20240410211143Z', 'creationDate': 'D:20240410211143Z', 'page': 3}, page_content='Scaled Dot-Product Attention \nMulti-Head Attention\nFigure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several\nattention layers running in parallel.\nof the values, where the weight assigned to each value is computed by a compatibility function of the\nquery with the corresponding key.\n3.2.1\nScaled Dot-Product Attention\nWe call our particular attention "Scaled Dot-Product Attention" (Figure 2). The input consists of\nqueries and keys of dimension dk, and values of dimension dv. We compute the dot products of the\nquery with all keys, divide each by √dk, and apply a softmax function to obtain the weights on the\nvalues.\nIn practice, we compute the attention function on a set of queries simultaneously, packed together\ninto a matrix Q. The keys and values are also packed together into matrices K and V . We compute\nthe matrix of outputs as:\nAttention(Q, K, V) = softmax(QKT\n√dk\n)V\n(1)\nThe two most commonly used attention functions are additive attention [2], and dot-product (multi-\nplicative) attention. Dot-product attention is identical to our algorithm, except for the scaling factor\nof \n1\n√dk . Additive attention computes the compatibility function using a feed-forward network with\na single hidden layer. While the two are similar in theoretical complexity, dot-product attention is\nmuch faster and more space-efficient in practice, since it can be implemented using highly optimized\nmatrix multiplication code.\nWhile for small values of dk the two mechanisms perform similarly, additive attention outperforms\ndot product attention without scaling for larger values of dk [3]. We suspect that for large values of\ndk, the dot products grow large in magnitude, pushing the softmax function into regions where it has\nextremely small gradients 4. To counteract this effect, we scale the dot products by\n1\n√dk .\n3.2.2\nMulti-Head Attention\nInstead of performing a single attention function with dmodel-dimensional keys, values and queries,\nwe found it beneficial to linearly project the queries, keys and values h times with different, learned\nlinear projections to dk, dk and dv dimensions, respectively. On each of these projected versions of\nqueries, keys and values we then perform the attention function in parallel, yielding dv-dimensional\n4To illustrate why the dot products get large, assume that the components of q and k are independent random\nvariables with mean 0 and variance 1. Then their dot product, q · k = Pdk\ni=1 qiki, has mean 0 and variance dk.\n4'),

Document(metadata={'producer': 'pdfTeX-1.40.25', 'creator': 'LaTeX with hyperref', 'creationdate': '2024-04-10T21:11:43+00:00', 'source': '..\\data\\pdf\\attention.pdf', 'file_path': '..\\data\\pdf\\attention.pdf', 'total_pages': 15, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2024-04-10T21:11:43+00:00', 'trapped': '', 'modDate': 'D:20240410211143Z', 'creationDate': 'D:20240410211143Z', 'page': 4}, page_content='output values. These are concatenated and once again projected, resulting in the final values, as\ndepicted in Figure 2. \nMulti-head attention allows the model to jointly attend to information from different representation\nsubspaces at different positions. With a single attention head, averaging inhibits this.\nMultiHead(Q, K, V) = Concat(head1, ..., headh)W O\nwhere headi = Attention(QW Q\ni , KW K\ni , V W V\ni)\nWhere the projections are parameter matrices W Q\ni\n∈ Rdmodel×dk,W K\ni \n∈ Rdmodel×dk,W V\ni\n∈ Rdmodel×dv\nand W O ∈ Rhdv×dmodel\nIn this work we employ h = 8 parallel attention layers, or heads. For each of these we use\ndk = dv = dmodel/h = 64. Due to the reduced dimension of each head, the total computational cost\nis similar to that of single-head attention with full dimensionality.\n3.2.3\nApplications of Attention in our Model\nThe Transformer uses multi-head attention in three different ways:\n• In "encoder-decoder attention" layers, the queries come from the previous decoder layer,\nand the memory keys and values

come from the output of the encoder. This allows every\nposition in the decoder to attend over all positions in the input sequence. This mimics the\ntypical encoder-decoder attention mechanisms in sequence-to-sequence models such as\n[38, 2, 9].\n• The encoder contains self-attention layers. In a self-attention layer all of the keys, values\nand queries come from the same place, in this case, the output of the previous layer in the\nencoder. Each position in the encoder can attend to all positions in the previous layer of the\nencoder.\n• Similarly, self-attention layers in the decoder allow each position in the decoder to attend to\nall positions in the decoder up to and including that position. We need to prevent leftward\ninformation flow in the decoder to preserve the auto-regressive property. We implement this\ninside of scaled dot-product attention by masking out (setting to −∞) all values in the input\nof the softmax which correspond to illegal connections. See Figure 2.\n3.3\nPosition-wise Feed-Forward Networks\nIn addition to attention sub-layers, each of the layers in our encoder and decoder contains a fully\nconnected feed-forward network, which is applied to each position separately and identically. This\nconsists of two linear transformations with a ReLU activation in between.\nFFN(x) = max(0, xW1 + b1)W2 + b2\n(2)\nWhile the linear transformations are the same across different positions, they use different parameters\nfrom layer to layer. Another way of describing this is as two convolutions with kernel size 1.\nThe dimensionality of input and output is dmodel = 512, and the inner-layer has dimensionality\ndff = 2048.\n3.4\nEmbeddings and Softmax\nSimilarly to other sequence transduction models, we use learned embeddings to convert the input\ntokens and output tokens to vectors of dimension dmodel. We also use the usual learned linear transfor-\nmation and softmax function to convert the decoder output to predicted next-token probabilities. In\nour model, we share the same weight matrix between the two embedding layers and the pre-softmax \nlinear transformation, similar to [30]. In the embedding layers, we multiply those weights by √dmodel.\n5'),

Document(metadata={'producer': 'pdfTeX-1.40.25', 'creator': 'LaTeX with hyperref', 'creationdate': '2024-04-10T21:11:43+00:00', 'source': '..\\data\\pdf\\attention.pdf', 'file_path': '..\\data\\pdf\\attention.pdf', 'total_pages': 15, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2024-04-10T21:11:43+00:00', 'trapped': '', 'modDate': 'D:20240410211143Z', 'creationDate': 'D:20240410211143Z', 'page': 5}, page_content='Table 1: Maximum path lengths, per-layer complexity and minimum number of sequential operations\nfor different layer types. n is the sequence length, d is the representation dimension, k is the kernel\nsize of convolutions and r the size of the neighborhood in restricted self-attention.\nLayer Type\nComplexity per Layer\nSequential\nMaximum Path Length\nOperations\nSelf-Attention\nO(n2 · d)\nO(1)\nO(1)\nRecurrent\nO(n · d2)\nO(n)\nO(n)\nConvolutional\nO(k · n · d2)\nO(1)\nO(logk(n))\nSelf-Attention (restricted)\nO(r · n · d)\nO(1)\nO(n/r)\n3.5\nPositional Encoding\nSince our model contains no recurrence and no convolution, in order for the model to make use of the\norder of the sequence, we must inject some information about the relative or absolute position of the\ntokens in the sequence. To this end, we add "positional encodings" to the input embeddings at the\nbottoms of the encoder and decoder stacks. The positional encodings have the same dimension dmodel\nas the embeddings, so that the two can be summed. There are many choices of positional encodings,\nlearned and fixed [9].\nIn this work, we use sine and cosine functions of different frequencies:\nPE(pos,2i) = sin(pos/100002i/dmodel)\nPE(pos,2i+1) = cos(pos/100002i/dmodel)\nwhere pos is the position and i is the dimension. That is, each dimension of the positional encoding\ncorresponds to a sinusoid. The wavelengths form a geometric progression from 2π to 10000 · 2π. We\nchose this function because we hypothesized it would allow the model to easily learn to attend by\nrelative positions, since for any fixed offset k, PEpos+k can be represented as a linear function of \nPEpos.\nWe also experimented with using learned positional embeddings [9] instead, and found that the two\nversions produced nearly identical results (see Table 3 row (E)). We chose the sinusoidal version\nbecause it may allow the model to extrapolate to sequence lengths longer than the ones encountered\nduring training.\n4\nWhy Self-Attention\nIn this section we compare various aspects of self-attention layers to the recurrent and convolu-\ntional layers commonly used for mapping one variable-length sequence of symbol representations\n(x1, ..., xn) to another sequence of equal length (z1, ..., zn), with xi, zi ∈ Rd, such as a hidden\nlayer in a typical sequence transduction encoder or decoder. Motivating our use of self-attention we

\nconsider three desiderata.\nOne is the total computational complexity per layer. Another is the amount of computation that can\nbe parallelized, as measured by the minimum number of sequential operations required.\nThe third is the path length between long-range dependencies in the network. Learning long-range\ndependencies is a key challenge in many sequence transduction tasks. One key factor affecting the\nability to learn such dependencies is the length of the paths forward and backward signals have to\ntraverse in the network. The shorter these paths between any combination of positions in the input\nand output sequences, the easier it is to learn long-range dependencies [12]. Hence we also compare\nthe maximum path length between any two input and output positions in networks composed of the\ndifferent layer types. \nAs noted in Table 1, a self-attention layer connects all positions with a constant number of sequentially\nexecuted operations, whereas a recurrent layer requires O(n) sequential operations. In terms of\ncomputational complexity, self-attention layers are faster than recurrent layers when the sequence\n6'),

Document(metadata={'producer': 'pdfTeX-1.40.25', 'creator': 'LaTeX with hyperref', 'creationdate': '2024-04-10T21:11:43+00:00', 'source': '..\\data\\pdf\\attention.pdf', 'file_path': '..\\data\\pdf\\attention.pdf', 'total_pages': 15, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2024-04-10T21:11:43+00:00', 'trapped': '', 'modDate': 'D:20240410211143Z', 'creationDate': 'D:20240410211143Z', 'page': 6}, page_content='length n is smaller than the representation dimensionality d, which is most often the case with\nsentence representations used by state-of-the-art models in machine translations, such as word-piece\n[38] and byte-pair [31] representations. To improve computational performance for tasks involving\nvery long sequences, self-attention could be restricted to considering only a neighborhood of size r in\nthe input sequence centered around the respective output position. This would increase the maximum\npath length to O(n/r). We plan to investigate this approach further in future work.\nA single convolutional layer with kernel width k < n does not connect all pairs of input and output \npositions. Doing so requires a stack of O(n/k) convolutional layers in the case of contiguous kernels,\nor O(logk(n)) in the case of dilated convolutions [18], increasing the length of the longest paths\nbetween any two positions in the network. Convolutional layers are generally more expensive than\nrecurrent layers, by a factor of k. Separable convolutions [6], however, decrease the complexity\nconsiderably, to O(k · n · d + n · d2). Even with k = n, however, the complexity of a separable\nconvolution is equal to the combination of a self-attention layer and a point-wise feed-forward layer,\nthe approach we take in our model.\nAs side benefit, self-attention could yield more interpretable models. We inspect attention distributions\nfrom our models and present and discuss examples in the appendix. Not only do individual attention \nheads clearly learn to perform different tasks, many appear to exhibit behavior related to the syntactic\nand semantic structure of the sentences.\n5\nTraining\nThis section describes the training regime for our models.\n5.1\nTraining Data and Batching\nWe trained on the standard WMT 2014 English-German dataset consisting of about 4.5 million\nsentence pairs. Sentences were encoded using byte-pair encoding [3], which has a shared source-\ntarget vocabulary of about 37000 tokens. For English-French, we used the significantly larger WMT\n2014 English-French dataset consisting of 36M sentences and split tokens into a 32000 word-piece \nvocabulary [38]. Sentence pairs were batched together by approximate sequence length. Each training\nbatch contained a set of sentence pairs containing approximately 25000 source tokens and 25000\ntarget tokens.\n5.2\nHardware and Schedule\nWe trained our models on one machine with 8 NVIDIA P100 GPUs. For our base models using\nthe hyperparameters described throughout the paper, each training step took about 0.4 seconds. We\ntrained the base models for a total of 100,000 steps or 12 hours. For our big models,(described on the\nbottom line of table 3), step time was 1.0 seconds. The big models were trained for 300,000 steps\n(3.5 days). \n5.3\nOptimizer\nWe used the Adam optimizer [20] with β1 = 0.9, β2 = 0.98 and □ = 10−9. We varied the learning\nrate over the course of training, according to the formula:\nlrate = d −0.5\nmodel · min(step_num−0.5, step_num · warmup_steps−1.5)\n(3)\nThis corresponds to increasing the learning rate linearly for the first warmup_steps training steps,\nand decreasing it thereafter proportionally to the inverse square root of the step number. We used\nwarmup_steps = 4000.\n5.4\nRegularization\nWe employ three types of regularization during training:\n7'),

Document(metadata={'producer': 'pdfTeX-1.40.25', 'creator': 'LaTeX with hyperref', 'creationdate': '2024-04-10T21:11:43+00:00', 'source': '..\\data\\pdf\\attention.pdf', 'file_path': '..\\data\\pdf\\attention.pdf', 'total_pages': 15, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2024-04-10T21:11:43+00:00', 'trapped': '', 'modDate': 'D:20240410211143Z', 'creationDate': 'D:20240410211143Z', 'page': 7}, page_content='Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the\nEnglish-to-German and English-to-French newstest2014 tests at a fraction of the training cost.\nModel\nBLEU \nTraining Cost (FLOPs)\nEN-DE\nEN-FR\nEN-DE\nEN-FR\nByteNet [18]\n23.75\nDeep-Att + PosUnk [39]\n39.2\n1.0 · 1020\nGNMT + RL [38]\n24.6\n39.92\n2.3 · 1019\n1.4 · 1020\nConvS2S [9]\n25.16\n40.46\n9.6 · 1018\n1.5 · 1020\nMoE [32]\n26.03\n40.56\n2.0 · 1019\n1.2 · 1020\nDeep-Att + PosUnk Ensemble [39]\n40.4\n8.0 · 1020\nGNMT + RL Ensemble [38]\n26.30\n41.16\n1.8 · 1020\n1.1 · 1021\nConvS2S Ensemble [9]\n26.36\n41.29\n7.7 · 1019\n1.2 · 1021\nTransformer (base model)\n27.3\n38.1\n3.3 · 1018\nTransformer (big)\n28.4\n41.8\n2.3 · 1019\nResidual Dropout\nWe apply dropout [33] to the output of each sub-layer, before it is added to the\nsub-layer input and normalized. In addition, we apply dropout to the sums of the embeddings and the\npositional encodings in both the encoder and decoder stacks. For the base model, we use a rate of\nPdrop = 0.1.\nLabel Smoothing\nDuring training, we employed label smoothing of value ls = 0.1 [36]. This\nhurts perplexity, as the model learns to be more unsure, but improves accuracy and BLEU score.\n6\nResults\n6.1\nMachine Translation\nOn the WMT 2014 English-to-German translation task, the big transformer model (Transformer (big)\nin Table 2) outperforms the best previously reported models (including ensembles) by more than 2.0\nBLEU, establishing a new state-of-the-art BLEU score of 28.4. The configuration of this model is\nlisted in the bottom line of Table 3. Training took 3.5 days on 8 P100 GPUs. Even our base model\nsurpasses all previously published models and ensembles, at a fraction of the training cost of any of\nthe competitive models.\nOn the WMT 2014 English-to-French translation task, our big model achieves a BLEU score of 41.0,\noutperforming all of the previously published single models, at less than 1/4 the training cost of the\nprevious state-of-the-art model. The Transformer (big) model trained for English-to-French used\ndropout rate Pdrop = 0.1, instead of 0.3.\nFor the base models, we used a single model obtained by averaging the last 5 checkpoints, which\nwere written at 10-minute intervals. For the big models, we averaged the last 20 checkpoints. We\nused beam search with a beam size of 4 and length penalty α = 0.6 [38]. These hyperparameters\nwere chosen after experimentation on the development set. We set the maximum output length during\ninference to input length + 50, but terminate early when possible [38].\nTable 2 summarizes our results and compares our translation quality and training costs to other model\narchitectures from the literature. We estimate the number of floating point operations used to train a\nmodel by multiplying the training time, the number of GPUs used, and an estimate of the sustained\nsingle-precision floating-point capacity of each GPU 5.\n6.2\nModel Variations\nTo evaluate the importance of different components of the Transformer, we varied our base model\nin different ways, measuring the change in performance on English-to-German translation on the\n5We used values of 2.8, 3.7, 6.0 and 9.5 TFLOPS for K80, K40, M40 and P100, respectively.\n8'),

Document(metadata={'producer': 'pdfTeX-1.40.25', 'creator': 'LaTeX with hyperref', 'creationdate': '2024-04-10T21:11:43+00:00', 'source': '..\\data\\pdf\\attention.pdf', 'file_path': '..\\data\\pdf\\attention.pdf', 'total_pages': 15, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2024-04-10T21:11:43+00:00', 'trapped': '', 'modDate': 'D:20240410211143Z', 'creationDate': 'D:20240410211143Z', 'page': 8}, page_content='Table 3: Variations on the Transformer architecture. Unlisted values are identical to those of the base\nmodel. All metrics are on the English-to-German translation development set, newstest2013. Listed\nperplexities are per-wordpiece, according to our byte-pair encoding, and should not be compared to\nper-word perplexities.\nN\ndmodel\ndff\nh\ndk\ndv\nPdrop\nls\ntrain\nPPL\nBLEU\nparams \nsteps\n(dev)\n(dev)\n×106\nbase\n6\n512\n2048\n8\n64\n64\n0.1\n0.1\n100K \n4.92\n25.8\n65\n(A)\n1\n512\n512\n5.29\n24.9\n4\n128\n128\n5.00\n25.5\n16\n32\n32\n4 .91\n25.8\n32\n16\n16\n5.01\n25.4\n(B)\n16\n5.16\n25.1\n58\n32\n5.01\n25.4\n60\n(C)\n2\

n6.11\n23.7\n36\n4\n5.19\n25.3\n50\n8\n4.88\n25.5\n80\n256\n32\n32\n5.75\n24.5\n28\n1024\n128\n128\n4.66\n26.0\n168\n1024\n5.12\n25.4\n53\n4096\n4.75\n26.2\n90\n(D)\n0.0\n5.77\n24.6\n0.2\n4.95\n25.5\n0.0\n4.67\n25.3\n0.2\n5.47\n25.7\n(E)\npositional embedding instead of sinusoids\n4.92\n25.7\nbig\n6\n1024\n4096\n16\n0.3\n300K \n4.33\n26.4\n213\ndevelopment set, newstest2013. We used beam search as described in the previous section, but no\ncheckpoint averaging. We present these results in Table 3.\nIn Table 3 rows (A), we vary the number of attention heads and the attention key and value dimensions, \nkeeping the amount of computation constant, as described in Section 3.2.2. While single-head \nattention is 0.9 BLEU worse than the best setting, quality also drops off with too many heads. \nIn Table 3 rows (B), we observe that reducing the attention key size dk hurts model quality. This \nsuggests that determining compatibility is not easy and that a more sophisticated compatibility\nfunction than dot product may be beneficial. We further observe in rows (C) and (D) that, as expected,\nbigger models are better, and dropout is very helpful in avoiding over-fitting. In row (E) we replace our\nsinusoidal positional encoding with learned positional embeddings [9], and observe nearly identical\nresults to the base model.\n6.3\nEnglish Constituency Parsing\nTo evaluate if the Transformer can generalize to other tasks we performed experiments on English\nconstituency parsing. This task presents specific challenges: the output is subject to strong structural\nconstraints and is significantly longer than the input. Furthermore, RNN sequence-to-sequence\nmodels have not been able to attain state-of-the-art results in small-data regimes [37].\nWe trained a 4-layer transformer with dmodel = 1024 on the Wall Street Journal (WSJ) portion of the\nPenn Treebank [25], about 40K training sentences. We also trained it in a semi-supervised setting,\nusing the larger high-confidence and BerkleyParser corpora from with approximately 17M sentences\n[37]. We used a vocabulary of 16K tokens for the WSJ only setting and a vocabulary of 32K tokens\nfor the semi-supervised setting.\nWe performed only a small number of experiments to select the dropout, both attention and residual\n(section 5.4), learning rates and beam size on the Section 22 development set, all other parameters\nremained unchanged from the English-to-German base translation model. During inference, we\n9'),

Document(metadata={'producer': 'pdfTeX-1.40.25', 'creator': 'LaTeX with hyperref', 'creationdate': '2024-04-10T21:11:43+00:00', 'source': '..\\data\\pdf\\attention.pdf', 'file_path': '..\\data\\pdf\\attention.pdf', 'total_pages': 15, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2024-04-10T21:11:43+00:00', 'trapped': '', 'modDate': 'D:20240410211143Z', 'creationDate': 'D:20240410211143Z', 'page': 9}, page_content='Table 4: The Transformer generalizes well to English constituency parsing (Results are on Section 23\nof WSJ)\nParser \nTraining\nWSJ 23 F1\nVinyals & Kaiser el al. (2014) [37]\nWSJ only, discriminative \n88.3\nPetrov et al. (2006) [29]\nWSJ only, discriminative\n90.4\nZhu et al. (2013) [40]\nWSJ only, discriminative\n90.4\nDyer et al. (2016) [8]\nWSJ only, discriminative\n91.7\nTransformer (4 layers)\nWSJ only, discriminative\n91.3\nZhu et al. (2013) [40]\nsemi-supervised \n91.3\nHuang & Harper (2009) [14]\nsemi-supervised\n91.3\nMcClosky et al. (2006) [26]\nsemi-supervised\n92.1\nVinyals & Kaiser el al. (2014) [37]\nsemi-supervised \n92.1\nTransformer (4 layers)\nsemi-supervised\n92.7\nLuong et al. (2015) [23]\nmulti-task \n93.0\nDyer et al. (2016) [8]\ngenerative\n93.3\nincreased the maximum output length to input length + 300. We used a beam size of 21 and α = 0.3\nfor both WSJ only and the semi-supervised setting.\nOur results in Table 4 show that despite the lack of task-specific tuning our model performs sur-\nprisingly well, yielding better results than all previously reported models with the exception of the\nRecurrent Neural Network Grammar [8].\nIn contrast to RNN sequence-to-sequence models [37], the Transformer outperforms the Berkeley-\nParser [29] even when training only on the WSJ training set of 40K sentences.\n7\nConclusion\nIn this work, we presented the Transformer, the first sequence transduction model based entirely on\nattention, replacing the recurrent layers most commonly used in encoder-decoder architectures with \nmulti-headed self-attention.\nFor translation tasks, the Transformer can be trained significantly faster than architectures based\non recurrent or convolutional layers. On both WMT 2014 English-to-German and WMT 2014\nEnglish-to-French translation tasks, we achieve a new state of the art. In the former task our best\nmodel outperforms even all previously reported ensembles.\nWe are excited about the future of attention-based models and plan to apply them

to other tasks. We\nplan to extend the Transformer to problems involving input and output modalities other than text and\nto investigate local, restricted attention mechanisms to efficiently handle large inputs and outputs\nsuch as images, audio and video. Making generation less sequential is another research goals of ours.\nThe code we used to train and evaluate our models is available at https://github.com/\ntensorflow/tensor2tensor.\nAcknowledgements \nWe are grateful to Nal Kalchbrenner and Stephan Gouws for their fruitful\ncomments, corrections and inspiration.\nReferences\n[1] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. arXiv preprint\narXiv:1607.06450, 2016.\n[2] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. Neural machine translation by jointly\nlearning to align and translate. CoRR, abs/1409.0473, 2014.\n[3] Denny Britz, Anna Goldie, Minh-Thang Luong, and Quoc V. Le. Massive exploration of neural\nmachine translation architectures. CoRR, abs/1703.03906, 2017.\n[4] Jianpeng Cheng, Li Dong, and Mirella Lapata. Long short-term memory-networks for machine\nreading. arXiv preprint arXiv:1601.06733, 2016.\n10'),

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attend to a distant dependency of\nthe verb ‘making’, completing the phrase ‘making...more difficult’. Attentions here shown only for\nthe word ‘making’. Different colors represent different heads. Best viewed in color.\n13'),

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Document(metadata={'producer': 'pikepdf 8.15.1', 'creator': 'arXiv GenPDF (tex2pdf:)', 'creationdate': '2025-09-01T00:50:53+00:00', 'source': '..\\data\\pdf\\emneddings.pdf', 'file_path': '..\\data\\pdf\\emneddings.pdf', 'total_pages': 27, 'format': 'PDF 1.4', 'title': 'QZhou-Embedding Technical Report', 'author': 'Peng Yu; En Xu; Bin Chen; Haibiao Chen; Yinfei Xu', 'subject': '', 'keywords': '', 'moddate': '2025-09-01T00:50:53+00:00', 'trapped': '', 'modDate': "D:20250901005053+00'00'", 'creationDate': "D:20250901005053+00'00'", 'page': 0}, page_content='QZhou-Embedding Technical Report\nKingsoft AI\nQZhou-Embedding Technical Report\nPeng Yu, En Xu, Bin Chen, Haibiao Chen, Yinfei Xu\nKingsoft AI\nAugust 2025\nAbstract\nWe present QZhou-Embedding, a general-purpose contextual text embed-\nding model with exceptional text representation capabilities.\nBuilt upon the\nQwen2.5-7B-Instruct foundation model, we designed a unified multi-task frame-\nwork comprising specialized data transformation and training strategies.\nThe\ndata transformation scheme enables the incorporation of more diverse textual\ntraining datasets, while the task-specific training strategies enhance model learn-\ning efficiency. We developed a data synthesis pipeline leveraging LLM API, in-\ncorporating techniques such as Paraphrasing, Augmentation, and Hard negative\nexample generation to improve the semantic richness and sample difficulty of\nthe training set. Additionally, we employ a two-stage training strategy, compris-\ning initial retrieval-focused pretraining followed by full-task fine-tuning, enabling\nthe embedding model to extend its capabilities based on robust retrieval perfor-\nmance. Our model achieves state-of-the-art

results on the MTEB and CMTEB\nbenchmarks, ranking first on both leaderboards(August 27, 2025), simultaneously\nachieves state-of-the-art performance on tasks including Reranking, Clustering,\netc. Our findings demonstrate that higher-quality, more diverse data is crucial for \nadvancing retrieval model performance, and that leveraging LLMs' generative ca-\npabilities can further optimize data quality for embedding model breakthroughs.\nOur model weights are released on HuggingFace1 under Apache 2.0 license. For\nreproducibility, we provide evaluation code and instructions on GitHub2.\n1\nIntroduction\nText embedding models, which transform natural language text into mathematical vec-\ntor representations, play an indispensable role in text mining, question-answering sys-\ntems, recommendation systems, and retrieval-augmented generation. Recently, LLM-\nbased agent technology has experienced rapid development and widespread adoption,\nembedding models, which transform textual or multimodal data into vector represen-\ntations for knowledge base construction, have significantly enhanced agent systems\n https://kingsoft.com/\n1https://huggingface.co/Kingsoft-LLM/QZhou-Embedding \n2https://github.com/Kingsoft-LLM/QZhou-Embedding\narXiv:2508.21632v1 [cs.CL] 29 Aug 2025'),

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work, we present QZhou-Embedding, built upon the powerful Qwen2.5-7B-\nInstruct[21] model, which pushes the boundaries of text embedding capabilities. To\nenhance the model's semantic understanding, we designed a unified multi-task learn-\ning framework that not only accommodates more diverse training data but also bring\nefficient learning across three key tasks: retrieval, natural language inference (NLI),\nand classification. Our framework comprises two core components: 1. Data Trans-\nformation: We carefully adapt data formats to the specific requirements of retrieval,\nNLI, and classification tasks, enabling e ective feature extraction from heterogeneous\ndata sources, significantly benefiting retrieval model training. 2. Training Strategy:\nWe designed specialized loss functions based on each task's characteristics, optimizing\nmodel training efficiency. To further improve the robustness and generalization of vec-\n2'),

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page_content='QZhou-Embedding Technical Report\nKingsoft AI\npretraining. As a result, LLM-based embeddings achieve superior performance in re-\ntrieval and related tasks. \n2.2\nEmbedding Model Training\nThe mainstream approaches currently involve contrastive learning pretraining on un-\nsupervised/weakly supervised corpora and supervised contrastive learning training on\nhigh-quality labeled positive and negative samples.\nIn unsupervised learning, early\nwork like SimCSE[7] proposed feeding continuous inputs of both original and noise-\naugmented texts while employing contrastive learning to enhance the model's dis-\ncriminative representation capability. For weakly supervised learning, gte[33] utilized\nlarge-scale structured data (web search data, title-article pairs, etc.) for pretraining,\nfollowed by fine-tuning on high-quality open-source retrieval training data, achieving\nperformance comparable to OpenAI embeddings with significantly fewer parameters.\nConan-Embedding[24] and v2 similarly adopted the weakly supervised pretraining &\nsupervised fine-tuning approach but incorporated techniques like cross-GPU batch loss\nbalancing, dynamic hard negative mining, and soft masking (v2) to optimize the model.\nSeed1.6-Embedding[38] employed a phased training strategy combining text and multi-\nmodal pretraining followed by business-scenario-specific fine-tuning, achieving superior\nrepresentation quality.\nSubstantial research has also been conducted on modeling di erent tasks. Piccolo2[39]\nintroduced multi-task hybrid loss functions for diverse downstream tasks, an approach\nwe also incorporate.\nSFR-Embedding[30] utilized multi-task learning techniques to\nregularize embeddings, significantly enhancing domain data discrimination. Xiaobu-\nembedding unified the treatment of major CMTEB problem categories from the per-\nspective of circle loss[40], fully leveraging multiple positive examples in original datasets\nwhile carefully balancing di erent loss weights. \n2.3\nData Synthesis\nData quantity and quality are the most critical factors in model optimization, data\nsynthesis methods have become a critical research direction due to the high cost of\nmanual annotation.\nDoc2Query[41] and Query2Doc[42] employ question-answering \nmodels to generate pseudo-queries and pseudo-documents respectively, enhancing data\nfor improved RAG performance.\nPromptagator[43] addresses few-shot retrieval sce-\nnarios by generating queries of diverse intents using few-shot demonstrations and an-\nnotations, e ectively improving retrieval capabilities across varying intents or distri-\nbutions.\nGPL[44] utilizes existing T5 encoder-decoder models to generate queries,\nretrieves similar passages as hard negatives using existing retrieval models, and em-\nploys cross-encoders to score each (query, passage) pair. Unnatural Instructions[45]\nleverages prompt and in-context learning (ICL) techniques to generate synthetic ex-\namples through controlled instructions, inputs, and constraints, producing 64k diverse\ndata entries from several seed examples with promising experimental results. Qwen3-\nEmbedding[34] designs a diversified prompting strategy by assigning document-specific\nroles to simulate potential users querying that document, enabling LLMs to generate\nstylistically authentic queries that enhance diversity and realism. \n4'),

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page_content='QZhou-Embedding Technical Report\nKingsoft AI\n2.4\nHard Negative Mining Techniques\nHard negatives serve as essential components in contrastive learning for retrieval model\ntraining. Early work like ANCE[46] proposed an asynchronous ANN indexing mech-\nanism that periodically updates hard negatives using checkpoint states to maintain\noptimally challenging samples. Conan-Embedding[24] and its v2 version implemented\na dynamic hard

negative sampling strategy by excluding and refreshing samples when\ntheir scores fall below a threshold. NV-Retriever[47] proposed positive-aware negative\nmining, introducing TopK-MarginPos and TopKPercPos filtering criteria to minimize\nfalse negatives. LGAI-Embedding[17] built upon NV-Retriever's strategy with adap-\ntive margin-based mining strategies, employing ANNA IR as a teacher retrieval model\nto identify high-quality hard negatives while using TopKPercPos filtering to eliminate\nfalse negatives.\n3\nUnified Multi-task Learning Framework \nEmbedding models support numerous downstream tasks including retrieval, reranking,\nSTS, and classification. Given the diversity of these tasks and their associated data\ncomplexity, we explore a unified strategy to e ectively handle them collectively while\npromoting optimization of the embedding model. Existing research on unified task pro-\ncessing includes circle loss[40], which approaches sentence pair similarity from a global\nperspective by categorizing tasks into class-level labels and pair-wise labels, Xiaobu-\nembedding demonstrated significant improvements by adopting this approach. Other\nmodels like Piccolo2[39], SFR-Embedding[30], NV-Embed[47], Conan-Embedding[24] ,\nand Conan-Embedding-v2 have incorporated multi-task learning using diverse train-\ning data with varying label processing methods, some employing task-specific losses\n(InfoNCE[48], Cosent[49], etc.).\nOur design principle aims to accommodate more tasks and data types, enabling cross-\ndomain and cross-task data to e ectively enhance embedding capabilities. We propose\na unified multi-task learning framework that categorizes training data into three task\ntypes: retrieval, NLI, and classification, with customized data and training solutions\nfor each, allowing most natural text data to be converted into embedding training data\nthrough this framework. The following sections detail the framework's components and\nimplementation methods.\n3.1\nModel Architecture \nEmbedding models based on BERT or T5 [39][15][50][24] exhibit powerful contextual \nrepresentation capabilities, primarily attributed to their bidirectional attention mech-\nanisms. However, recent large language models predominantly adopt decoder-only ar-\nchitectures with unidirectional attention, significantly constraining tokens' ability to\ncapture contextual information. Several studies have addressed this limitation through\narchitectural modifications or attention mechanism optimizations[12][31][47]. Our work\nbuilds upon the Qwen2.5-7B-Instruct architecture and checkpoint due to its exceptional\nChinese language contextual capabilities. Consequently, we implemented the following\nmodifications: (1) modifying the original causal attention to bi-directional attention\n5'),

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each evidence instance labeled as ”Supports”\nor ”Refutes”. The transformation process involves: converting the claim portion\n6'),

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designed specialized training objectives to\nto enhance model training efficiency.\nThis section elaborates on the design of loss\nfunctions for retrieval, NLI, and classification tasks. \n3.3.1\nRetrieval\nFor the retrieval task, we adopt the widely used InfoNCE loss[48], but incorporate an\nimprovement inspired by gte[33] by augmenting the original query-negative loss with an\nadditional query-query loss term. Specifically, each query within a batch is treated as a \nnegative sample for all other queries. The final loss formulation is explicitly described\nin Equation (1).\nLRetrieval = −1\nn\nX\ni\nlog\nesim(qi,d+\ni)/τ\nesim(qi,d+\ni)/τ + P\nj esim(qi,d−\nj)/τ + P\nj̸=iesim(qi,qj)/τ\n(1)\n3.3.2\nNLI\nFor NLI tasks, the transformed labels are numerically comparable and exhibit ordinal\nrelationships.\nWe employ Cosent loss[49] to optimize such data, which is designed\nbased on the principles of Circle loss[40]. As a ranking-sensitive loss function, Cosent\nloss requires only ordinal label information for optimization while demonstrating faster\nconvergence. Its mathematical formulation is presented in Equation (2).\n8'),

Document(metadata={'producer': 'pikepdf 8.15.1', 'creator': 'arXiv GenPDF (tex2pdf:)', 'creationdate': '2025-09-01T00:50:53+00:00', 'source': '..\\data\\pdf\\emneddings.pdf', 'file_path': '..\\data\\pdf\\emneddings.pdf', 'total_pages': 27, 'format': 'PDF 1.4', 'title': 'QZhou-Embedding Technical Report', 'author': 'Peng Yu; En Xu; Bin Chen; Haibiao Chen; Yinfei Xu', 'subject': '', 'keywords': '', 'moddate': '2025-09-01T00:50:53+00:00', 'trapped': '', 'modDate': "D:20250901005053+00'00'", 'creationDate': "D:20250901005053+00'00'", 'page': 8}, page_content='QZhou-Embedding Technical Report\nKingsoft AI\nLNLI = log(1 +\nX \nsim(i,j)>sim(k,l)\nexp(sim(xk, xl) −sim(xi, xj)\nτ\n))\n(2)\n3.3.3\nCLS\nThe classification loss also adopts the InfoNCE objective. However, since CLS data is\nprocessed in an example-based manner, directly applying in-batch negative sampling\non classification datasets with limited categories may lead to false negatives from items\nof di erent classes.\nNumerous studies have proposed diverse approaches to address\nthis issue[51][52][47]. We propose a masking mechanism that appends class labels to\neach positive and negative sample during preprocessing (recorded as separate variables\nrather than modifying raw text). During in-batch negative sampling, for each negative\nsample from other data instances, we check whether its label matches the current query's\nclass. If matched, the negative loss contribution is masked to zero to prevent erroneous\npenalization; otherwise, it is normally computed. The core loss remains InfoNCE, with\nthe CLS loss formulation shown in Equation (3). Where Cti denotes the class label of\nsample ti, and nrepresents the number of negative samples per data instance. \nLCLS = −1\nn\nX\ni\nlog esim(ti,t+\ni)/τ\nZi\n(3)\nwhere Zi = esim(ti,t+\ni)/τ +\nX\nn \nMASK(ti, t−\ni,n) · esim(ti,t−\ni,n)/τ+\nX\nj̸=iMASK(ti, tj) · esim(ti,tj)/τ+\nX\nj̸=iX\nn \nMASK(ti, t−\nj,n) · esim(ti,t−\nj,n)/τ\nand Cti = Ct+\ni\nand MASK(ti, tj) =\n(\n0\nif Cti = Ctj, \n1\notherwise\n4\nData Synthesis\nThe production of higher-quality data through data production has gained critical im-\nportance in embedding training.\nManual annotation incurs higher costs and lower\nproduction efficiency, thus developing e ective automated data synthesis methods has\nemerged as a key research focus. Recent advancements in large language models (LLMs)\nhave significantly improved their linguistic capabilities, enabling accurate interpretation\nof human instructions and generation of high-quality outputs. Multiple existing meth-\nods have e ectively leveraged LLMs to generate high-quality data[28][34], we similarly\n9'),

Document(metadata={'producer': 'pikepdf 8.15.1', 'creator': 'arXiv GenPDF (tex2pdf:)', 'creationdate': '2025-09-01T00:50:53+00:00', 'source': '..\\data\\pdf\\emneddings.pdf', 'file_path': '..\\data\\pdf\\emneddings.pdf', 'total_pages': 27, 'format': 'PDF 1.4', 'title': 'QZhou-Embedding Technical Report', 'author': 'Peng Yu; En Xu; Bin Chen; Haibiao Chen; Yinfei Xu', 'subject': '', 'keywords': '', 'moddate': '2025-09-01T00:50:53+00:00', 'trapped': '', 'modDate': "D:20250901005053+00'00'", 'creationDate': "D:20250901005053+00'00'", 'page': 9}, page_content='QZhou-Embedding Technical Report\nKingsoft AI\nleverages LLM capabilities for data production across three dimensions: structural di-\nversity, semantic diversity, and difficulty, with dedicated synthesis strategies for each.\nFor structural diversity, we propose Paraphrasing techniques; for semantic diversity,\nwe introduce Augmentation methods; and to increase

training difficulty and improve\nsemantic discriminability, we employ LLMs to generate more challenging hard negative\nexamples. The following sections detail these methodologies. The constraint compo-\nnents for all data synthesis techniques are specified in Table 5 of Appendix A.1.\n4.1\nStructural Diversity Enhancement\nLinguistic structures of text encompass lexical, syntactic, and grammatical features,\nwhich represent relatively surface-level characteristics reflecting word arrangements,\ncombinations, tenses, voices, and other formal attributes. \nEmbedding models must\naccurately capture underlying semantics despite variations in surface form, ensuring\nrobustness to external structural changes. For example, the following two sentences,\ndespite structural di erences, should be recognized as semantically equivalent: \n• The cat chased the mouse.\n• The mouse was chased by the cat.\nTo e ectively train an embedding model that remains invariant to structural variations\nwhile accurately capturing semantic information, we propose a Paraphrasing strategy.\nFor each training sample containing a query and a positive document, we apply LLM-\nbased paraphrasing to both contents, generating augmented instances that preserve\nsemantic equivalence while introducing structural divergence. The prompt constraints\nand workflow are illustrated in Figure 3.\nFigure 3: LLM-based Paraphrasing Workflow\n4.2\nSemantic Diversity Enhancement\nMerely augmenting data through superficial structural modifications yields negligible\nimprovements in model capabilities, as generalization relies not only on structural dis-\nentanglement but also on diverse topics and content to ensure uniform vector rep-\nresentations in the spatial domain. Therefore, beyond paraphrasing, we propose an\naugmentation method using LLM to diversify semantics. The core concept is: given a\n10'),

Document(metadata={'producer': 'pikepdf 8.15.1', 'creator': 'arXiv GenPDF (tex2pdf:)', 'creationdate': '2025-09-01T00:50:53+00:00', 'source': '..\\data\\pdf\\emneddings.pdf', 'file_path': '..\\data\\pdf\\emneddings.pdf', 'total_pages': 27, 'format': 'PDF 1.4', 'title': 'QZhou-Embedding Technical Report', 'author': 'Peng Yu; En Xu; Bin Chen; Haibiao Chen; Yinfei Xu', 'subject': '', 'keywords': '', 'moddate': '2025-09-01T00:50:53+00:00', 'trapped': '', 'modDate': "D:20250901005053+00'00'", 'creationDate': "D:20250901005053+00'00'", 'page': 10}, page_content='QZhou-Embedding Technical Report\nKingsoft AI\ncomplete (query, positive) pair, the model must comprehend the domain and perspec-\ntive discussed and learn to expand into di erent topics, aspects, and viewpoints while\nremaining contextually anchored. This process is governed via prompt constraints. The\nAugmentation framework is illustrated in Figure 4.\nFigure 4: Semantic Augmentation Workflow\nFigure 5: Hard Negative Synthesis Workflow\n4.3\nMore challenging embeddings\nHard negative examples are crucial for enhancing the performance of text embedding\nmodels, often requiring substantial e ort to acquire. Leveraging the linguistic capabili-\nties of large language models, we design an automated hard negative synthesis method\ntailored for retrieval datasets. Our domain-specific experiments demonstrate that large\nlanguage models can generate examples that are indistinguishable, the framework is\nillustrated in Figure 5.\nDuring Data paraphrasing and Augmentation, we implement task-specific strategies:\nfor retrieval tasks, we rewrite/expand (query, positive) pairs and add them to the orig-\ninal dataset; for NLI tasks, we rewrite individual sentences by randomly duplicating\nexisting entries containing the original sentences and replacing them with rewritten\nversions to achieve data expansion—without applying augmentation to prevent ambi-\nguity; for classification tasks, we rewrite sentences while retaining their original labels,\nexample-based processing was applied using the rewritten results, again without em-\nploying augmentation. We provide several data synthesis examples in Appendix A.3\nfor reference.\n11'),

Document(metadata={'producer': 'pikepdf 8.15.1', 'creator': 'arXiv GenPDF (tex2pdf:)', 'creationdate': '2025-09-01T00:50:53+00:00', 'source': '..\\data\\pdf\\emneddings.pdf', 'file_path': '..\\data\\pdf\\emneddings.pdf', 'total_pages': 27, 'format': 'PDF 1.4', 'title': 'QZhou-Embedding Technical Report', 'author': 'Peng Yu; En Xu; Bin Chen; Haibiao Chen; Yinfei Xu', 'subject': '', 'keywords': '', 'moddate': '2025-09-01T00:50:53+00:00', 'trapped': '', 'modDate': "D:20250901005053+00'00'", 'creationDate': "D:20250901005053+00'00'", 'page': 11}, page_content='QZhou-Embedding Technical Report\nKingsoft AI\nFigure 6: Training pipeline

\n5\nTraining Optimization\n5.1\nData Grouping Strategy\nPrior works like Linq-Embedding[52] and SFR-Embedding-Mistral[30] adopted task-\nhomogeneous batching, partitioning data by task rather than mixing them, and sam-\npling tasks based on weighted randomness during training. Building on this, we propose\na refined Data Grouping Strategy, extending the granularity from task-level to dataset-\nlevel partitioning. We posit that dataset-level grouping captures more domain-specific\nclustering patterns—samples within the same dataset often exhibit inherent domain\nsimilarities, while such consistency may not hold across datasets.\nOur approach partitions training data into subsets by name. During training, only\nsamples from a single dataset are sampled per batch, with file pointers recorded to\nenable sequential reading in subsequent iterations. For sampling weights, we adopt\nthe data sampling strategy from gte[33] and mgte[50], scaling weights by dataset size\nfollowed by normalization. For dataset i with size li, its sampling weight is computed\nas Equation (4)\npi =\nlα\ni\nPm\nj=1 lα\nj \n(4)\n5.2\nTwo-Stage Training\nInspired by NV-Embed's[47] two-stage contrastive learning instruction tuning tech-\nnique, we adopt a similar training approach: the first stage exclusively uses retrieval-\noriented training data, while the second stage integrates both retrieval and non-retrieval\ntasks, the overall training framework is illustrated in the figure 6. Two key distinctions \nare incorporated: first, we integrate the previously described Data Grouping Strat-\negy; second, we implement global control over the sampling ratio of retrieval training\ndatasets, since our findings indicate that naively incorporating additional data signifi-\ncantly degrades retrieval performance.\nFor global control of sampling ratio, a hyperparameter η is introduced into the sampling\n12'),

Document(metadata={'producer': 'pikepdf 8.15.1', 'creator': 'arXiv GenPDF (tex2pdf:)', 'creationdate': '2025-09-01T00:50:53+00:00', 'source': '..\\data\\pdf\\emneddings.pdf', 'file_path': '..\\data\\pdf\\emneddings.pdf', 'total_pages': 27, 'format': 'PDF 1.4', 'title': 'QZhou-Embedding Technical Report', 'author': 'Peng Yu; En Xu; Bin Chen; Haibiao Chen; Yinfei Xu', 'subject': '', 'keywords': '', 'moddate': '2025-09-01T00:50:53+00:00', 'trapped': '', 'modDate': "D:20250901005053+00'00'", 'creationDate': "D:20250901005053+00'00'", 'page': 12}, page_content='QZhou-Embedding Technical Report\nKingsoft AI\nfunction to control the proportion of retrieval training, ensuring that throughout the\nsecond training stage, the computational contribution of retrieval data accounts for η,\nwhile non-retrieval data constitutes 1−η. The following set of equations formalizes the\ncomputational process from partitioned datasets to sampling ratio determination. Let\nthe training data D = [d1, d2, ..., dN] , where each di represents a distinct dataset (e.g.,\nMSMARCO passage, SQUAD), with corresponding sizes L = [l1, l2, ..., lN]. Following\nthe aforementioned strategy, we first apply an exponential scaling factor α, a mask fac-\ntor M is then applied to filter retrieval and non-retrieval training sets for summation.\nThe equations are as follows:\nSret =\nX\ni\nMi · lα\ni\nSnon ret =\nX\ni\n(1 −Mi) · lα\ni\nwhere Mi =\n(\n0\nif di ∈ RET,\n1\nelse\nwhere RET denotes the set of retrieval training datasets. The retrieval ratio is then\nscaled using η to derive the final normalized sampling ratios for the training sets:\nLsamp = [lsamp\n1\n, lsamp\n2\n, ...lsamp\nN\n]\nwhere lsamp\ni\n= \n(ηRET ·lα\ni\nSret\nif di ∈ RET,\n(1−ηRET)·lα\ni\nSnon ret\nelse\n6\nExperiments \n6.1\nTraining Dataset\nPrimary data sources include bge-en-icl, bge-m3-data, and bge-multilingual-gemma2-\ndata 3 . The E5 dataset (approximately 1.5M samples) 4, utilized in E5-Mistral-7B[28],\nEcho Embedding[11], and LLM2Vec[12], is also incorporated.\nThe aforementioned\ndatasets include commonly used retrieval training corpora such as MS MARCO (both\npassage and document versions)[64], Natural Questions (NQ)[65], ELI5[66], HotpotQA[67], \nMIRACL[68], SQuAD[69], FEVER[70], Quora Question Pairs(QQP), and DuReader[71],\netc. \nPrevious researchers have already systematically collected and organized these\ndatasets, making them readily usable, we solely utilized the proposed method to update\nharder negative samples. Stella's[53] retrieval data llm 5 provides high-quality (query,\npositive, negative) triplets, while zpoint leverages datasets such as Huatuo medical QA6,\nall above data has been incorporated. Additional data from huggingface's sentence-\ntransformers7 repository includes reddit, hover[72], mr-tydi[73], law-gpt, and s2orc[74].\n3https://github.com/FlagOpen/FlagEmbedding/tree/master/dataset\n4https://drive.google.com/file/d/1YqgaJIzmBIH37XBxpRPCVzV CLh6aOI4/view\n5https://huggingface.co/datasets/infgrad/retrieval data llm\n6https://huggingface.co/iampanda/zpoint large embedding zh\n7https://

huggingface.co/sentence-transformers\n13'),

Document(metadata={'producer': 'pikepdf 8.15.1', 'creator': 'arXiv GenPDF (tex2pdf:)', 'creationdate': '2025-09-01T00:50:53+00:00', 'source': '..\\data\\pdf\\emneddings.pdf', 'file_path': '..\\data\\pdf\\emneddings.pdf', 'total_pages': 27, 'format': 'PDF 1.4', 'title': 'QZhou-Embedding Technical Report', 'author': 'Peng Yu; En Xu; Bin Chen; Haibiao Chen; Yinfei Xu', 'subject': '', 'keywords': '', 'moddate': '2025-09-01T00:50:53+00:00', 'trapped': '', 'modDate': "D:20250901005053+00'00'", 'creationDate': "D:20250901005053+00'00'", 'page': 13}, page_content='QZhou-Embedding Technical Report\nKingsoft AI\nOther sources encompass web questions, BioASQ[54], cmrc[55], CSL8, nli for simcse\n(used in SimCSE[7] and GTE[33]), MLDR9, GLUE Benchmark[56], Yelp Reviews[57]\nand Weibo Sentiment10 training sets.\nWe further integrate MTEB evaluation-related datasets like Imdb-Classification[58],\nMassiveIntent-Classification[59], MassiveScenario-Classification[59], STS12[60], LCQMC[61],\nPAWSX[62], and STSB[63], we utilized the training split from these datasets with con-\ntamination exclusion applied to remove samples highly similar to test sets.\nFor data requiring format conversion, we apply the methodologies described in Sen-\ntion 3.2.\nDatasets with limited samples (e.g., subsets of bge and e5 series, Imdb-\nClassification, STS12, LCQMC) are augmented via Paraphrasing and Augmentation\n(typically applied to datasets with fewer than 60k samples), we ultimately obtained ap-\nproximately 5M high-quality training samples through API interfaces. We deduplicate\nall training sets and filter out samples with low query-pos scores using GTE-Qwen2-7B-\nInstruct 11. For retrieval data lacking hard negatives, we employ synthetic hard negative\ngeneration. Due to API cost constraints, only 30% of hard negatives are synthetically \ngenerated; the remainder are produced using stella-large-zh-v3-1792d[53], with top-10\nto top-30 ranked results selected as hard negatives. The final training dataset contains\n11M quadruples (query, pos, neg, instruction) in total.\n6.2\nTrainset Instructions\nFor most training data containing instruction formats, we retain their original con-\ntents. For the MTEB training set, we adopt instructions corresponding to its evalu-\nation(consistent with Qwen3-Embedding runtime). For external data lacking instruc-\ntions (e.g., Huatuo, Reddit, Law-GPT, GLUE), we design task-specific and domain-\nadaptive instructions. Partial instruction templates are provided in Appendix A.2.\n6.3\nTraining Details\nAs previously mentioned, we adopt a two-stage training approach. For the first-stage\nretrieval training, we train on all retrieval datasets, with a warm-up step of 300 and\na learning rate of 3e-5, the total step of training is 32k. In the second stage, we use\nall training data, set the learning rate to 2e-5, and train for 8k steps, keeping all other\nconfigurations the same as in the first stage. We employ a batch size of 256 for all data\nusing the InfoNCE loss (i.e., retrieval and classification), considering data using the \ncosent loss (i.e., NLI), due to lower memory consumption from the absence of forward \ncomputation for negative samples, the batch size is set to 768. Across all stages, we\nemploy bfloat16 precision, with 4 hard negative samples and a cosine temperature of\n0.02, using Adam optimizer with a weight decay of 0.01. The Data Grouping Strategy\nremains unchanged between the two stages, except that the second stage incorporates\nall data with a global retrieval ratio ηRET of 0.72. Unlike existing works that commonly\n8https://github.com/ydli-ai/CSL?tab=readme-ov-file\n9https://huggingface.co/datasets/Shitao/MLDR\n10https://github.com/SophonPlus/ChineseNlpCorpus?tab=readme-ov-file\n11https://huggingface.co/Alibaba-NLP/gte-Qwen2-7B-instruct\n14'),

Document(metadata={'producer': 'pikepdf 8.15.1', 'creator': 'arXiv GenPDF (tex2pdf:)', 'creationdate': '2025-09-01T00:50:53+00:00', 'source': '..\\data\\pdf\\emneddings.pdf', 'file_path': '..\\data\\pdf\\emneddings.pdf', 'total_pages': 27, 'format': 'PDF 1.4', 'title': 'QZhou-Embedding Technical Report', 'author': 'Peng Yu; En Xu; Bin Chen; Haibiao Chen; Yinfei Xu', 'subject': '', 'keywords': '', 'moddate': '2025-09-01T00:50:53+00:00', 'trapped': '', 'modDate': "D:20250901005053+00'00'", 'creationDate': "D:20250901005053+00'00'", 'page': 14}, page_content='QZhou-Embedding Technical Report\nKingsoft AI\nuse LoRA fine-tuning, we employ full-parameter fine-tuning at all stages to ensure\nmaximum performance improvement. The query and passage lengths are set to 256\nand 1536 respectively. However, in practice, the model can handle sequences up to 8k\nin length due to the strong length extrapolation capability

of the RoPE[35] positional\nencoding used in most LLMs. The hyperparameter configurations for all training stages\nare provided in the table 1.\nTable 1: Training Hyperparameter Specifications \nItem\nStage1\nStage2\nWarm-up\n300\nSteps\n3e-5\n2e-5\nLR\n32k\n8k\nBatch Size InfoNCE\n256\nBatch Size Cosent\n-\n768\nPrecision\nbfloat16\nTemperature \n0.02\nOptimizer\nAdam\nQuery Length\n256\nPassage Length\n1536\n6.4\nCompared Methods\nWe selected the top-10 ranked models(August 27, 2025) on the MTEB/CMTEB leader-\nboards prior to the release of QZhou-Embedding as baselines. For MTEB, the compar-\nnative models include LGAI-Embedding-Preview[17], the Seed series (v1.5[75] , v1.6[38]),\nQwen series (8B, 4B)[34], ritrieve zh v1, xiaobu-embedding-v2, gemini-embedding-001[76],\njasper en vision language v1[14], Linq-Embed-Mistral[52], SFR-Embedding-Mistral[30],\nand NV-Embed-v2[47]. For CMTEB, the baseline models comprise the Seed series (as\nabove), Qwen series (as above), Conan series (v1[24], v2[13]), zpoint large embedding zh,\nand piccolo-large-zh-v2[39]. \n6.5\nMain Results\nThis section presents the evaluation results of Qzhou-embedding on MTEB/CMTEB\nbenchmarks, alongside comparative scores from the top 10 ranked models. As detailed\nin Table 2, Table 3, Qzhou-embedding achieves state-of-the-art performance across \nboth task-level and task-type average metrics, demonstrating the e ectiveness of our \napproach.\nFurthermore, under MTEB's official ranking protocol, Qzhou-embedding\nsecured the top position on both leaderboards. (Note: Highlighted maximum values\nin certain columns may reflect the best performance among the listed models rather\nthan the overall leaderboard maximum, as exemplified by the MTEB/classification\nbenchmark where the top score does not appear in the top 10 models.)\n15'),

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embedding model with exceptional text representation capabilities. We designed a\nunified multi-task framework comprising specialized data transformation and training\nstrategies, e ectively enhanced the diversity of training data. To further improve the\nquality of training data and the model's generalization capabilities, we developed a data\nsynthesis pipeline leveraging LLM API, incorporating techniques such as Paraphrasing,\nAugmentation, and Hard negative example generation. We employ a two-stage training\nstrategy comprising initial retrieval-focused training followed by full-task fine-tuning,\nenabling the embedding model to extend its capabilities based on robust retrieval per-\nformance.\nThe model achieves state-of-the-art results on the MTEB and CMTEB\nbenchmarks, ranking first on both leaderboards. Our findings establish that data qual-\nity and diversity are pivotal for improving embedding model capabilities. In the future,\nwe will focus on developing multimodal and multilingual embedding models, as well\nas exploring e ective applications of embedding models in agent systems, aiming to\nintegrate cutting-edge technologies to optimize this classical module.\nReferences \n[1] Robertson, Stephen E., and Steve Walker. "Some simple e ective approximations to\nthe 2-poisson model for probabilistic weighted retrieval." In SIGIR'94: Proceedings\n16'),

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Conference\non Empirical Methods in Natural Language Processing, pp. 2369–2380, Brussels, \nBelgium, October-November 2018. Association for Computational Linguistics. doi:\n10.18653/v1/D18-1259. URL https://aclanthology.org/D18-1259.\n[68] Xinyu Zhang, Nandan Thakur, Odunayo Ogundepo, Ehsan Kamalloo, David\nAlfonso-Hermelo, Xiaoguang Li, Qun Liu, Mehdi Rezagholizadeh, and Jimmy Lin.\nMiracl: A multilingual retrieval dataset covering 18 diverse languages. Transactions\nof the Association for Computational Linguistics, 11:1114–1131, 2023.\n[69] Pranav\nRajpurkar,\nJian\nZhang,\nKonstantin\nLopyrev,\nand\nPercy\nLiang. \nSquad:\n100,000+ questions for machine comprehension of text. arXiv preprint \narXiv:1606.05250, 2016.\n[70] James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mit-\ntal. Fever: a large-scale dataset for fact extraction and verification. arXiv preprint \narXiv:1803.05355, 2018.\n[71] Wei He, Kai Liu, Jing Liu, Yajuan Lyu, Shiqi Zhao, Xinyan Xiao, Yuan Liu,\nYizhong Wang, Hua Wu, Qiaoqiao She, Xuan Liu, Tian Wu, and Haifeng Wang.\n2018. DuReader: a Chinese Machine Reading Comprehension Dataset from Real-\nworld Applications. In Proceedings of the Workshop on Machine Reading for Ques-\ntion Answering, pages 37–46, Melbourne, Australia. Association for Computational\nLinguistics.\n[72] Yichen Jiang, Shikha Bordia, Zheng Zhong, Charles Dognin, Maneesh Singh, and\nMohit Bansal. 2020. HoVer: A Dataset for Many-Hop Fact Extraction And Claim\nVerification. In Findings of the Association for Computational Linguistics: EMNLP\n2020, pages 3441–3460, Online. Association for Computational Linguistics.\n[73] Zhang X, Ma X, Shi P, et al. Mr. TyDi: A multi-lingual benchmark for dense\nretrieval[J]. arXiv preprint arXiv:2108.08787, 2021.\n[74] Kyle Lo, Lucy Lu Wang, Mark Neumann, Rodney Kinney, and Daniel Weld. 2020.\nS2ORC: The Semantic Scholar Open Research Corpus. In Proceedings of the 58th\nAnnual Meeting of the Association for Computational Linguistics, pages 4969–4983,\nOnline. Association for Computational Linguistics.\n22'),

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page_content='QZhou-Embedding Technical Report\nKingsoft AI\nA.2\nInstruction Examples \nTable 5: Instruction for partial training data\nDataset\nInstruction\nHuatuo\nGiven a medical question, retrieve user replies that\nbest answer the question\nReddit\nRetrieve the paragraph most semantically similar\nto the given statement\nLaw-GPT\nRetrieve relevant legal provisions or interpreta-\ntions for the given case\nMNLI/SNLI\nRetrieve semantically similar text\nYelp \nClassify the customer review of businesses\nWeibo\nClassify the sentiment of Weibo comments\nA.3\nData Synthesis Examples\nNote: The text highlighted in yellow represents the original sentence, followed by the\nsynthetically generated sentence.\nTable 6: Paraphrasing Example (1)\nquery\npos\nWhat is the best credit\ncard for someone with no\ncredit history? \nIf you've never had a credit card before a likely\nreason can be due to lack of credit history. You\ncan apply for a department store card.\nWhat's the ideal credit\ncard for a person without \nany credit history?\nIf you've never had a credit card, it's probably\nbecause you don't have a credit history. A depart-\nment store card could be a good option to apply\nfor.\nWhat's the top credit card\nchoice for someone who has\nno credit history?\nIf you've never owned a credit card, it's probably\nbecause you don't have a credit history. A depart-\nment store card might be a good option to con-\nsider.\n24'),

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page_content='QZhou-Embedding Technical Report\nKingsoft AI\nTable 7: Paraphrasing Example (2)\nquery\npos\nWhich English Poet\nLaureate wrote 'The Faerie\nQueene'?\nEnglish Renaissance to begin, shakily, in the 1520s,\nand it continued until perhaps 1620. England had \na strong tradition of literature in the English ver-\nnacular, which gradually increased as English use\nof the printing press became common during the\nmid 16th century. By the time of Elizabethan liter-\nature a vigorous literary culture in both drama and\npoetry included poets such as Edmund Spenser,\nwhose verse epic 'The Faerie Queene' had a strong\ninfluence on English literature but was eventu-\nally overshadowed by the lyrics of William Shake-\nspeare, Thomas Wyatt and others. Typically, the\nworks of these playwrights and poets circulated in \nmanuscript form.\nWho was the English Poet\nLaureate that penned 'The\nFaerie Queene'? \nThe English Renaissance began, albeit slowly, in\nthe 1520s and likely lasted until around 1620. Eng-\nland boasted a robust literary tradition in the En-\nglish language, which expanded as the printing\npress became more widely adopted during the mid-\n16th century. By the time of Elizabethan litera-\nture, a dynamic literary scene in both drama and\npoetry emerged, featuring writers like Edmund\nSpenser, whose epic poem The Faerie Queene sig-\nnificantly influenced English literature, although\nit was eventually overshadowed by the works of\nWilliam Shakespeare, Thomas Wyatt, and others.\nGenerally, the writings of these poets and play-\nwrights circulated in manuscript form.\nWhich English Poet\nLaureate authored 'The\nFaerie Queene'?\nThe English Renaissance began somewhat uncer-\ntainly in the 1520s, continuing through to around\n1620. England had a rich literary heritage in its\nnative tongue, which expanded as the printing\npress became more widely used in the mid-16th\ncentury.\nBy the Elizabethan era, a thriving lit-\nerary culture in both drama and poetry emerged,\nwith poets like Edmund Spenser, whose epic The\nFaerie Queene profoundly impacted English liter-\nature, though it was eventually eclipsed by the\nworks of William Shakespeare, Thomas Wyatt,\nand others. Generally, the writings of these play-\nwrights and poets were circulated in manuscript

\nform.\n25'),

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page_content='QZhou-Embedding Technical Report\nKingsoft AI\nTable 8: Augmentation Example\nquery\npos\nneg\nWhat is the best\ncredit card for\nsomeone with no\ncredit history?\nIf you've never had a credit\ncard before a likely reason\ncan be due to lack of credit \nhistory. You can apply for a\ndepartment store card.\n-\nWhich credit card is\neasiest to get \napproved for with bad\ncredit?\nFor those with poor credit\nscores, secured credit cards \nfrom Capital One or Dis-\ncover are often the most ac-\ncessible options since they\nrequire \na\nrefundable\nde-\nposit.\nSome premium travel cards\nlike Chase Sapphire have\ngreat rewards but usually\nneed excellent credit. Store\ncards\nmay\naccept\nlower\nscores but have high inter-\nest rates.\nWhat credit cards\ncan I get as a college\nstudent with no\ncredit? \nStudents without credit his-\ntory often qualify for starter\ncards like Discover it Stu-\ndent or Capital One Jour-\nney, which are designed to\nhelp build credit.\nPremium cards like Amer-\nican Express Platinum re-\nquire good credit history,\nthough some banks\no er\nstudent accounts with debit\ncards.\nWhich English Poet\nLaureate wrote 'The\nFaerie Queene'?\n...By \nthe\ntime\nof\nEliz-\nabethan literature a vig-\norous\nliterary\nculture\nin\nboth drama and poetry in-\ncluded poets such as Ed-\nmund Spenser, whose verse\nepic 'The Faerie Queene'\nhad a strong influence on\nEnglish literature but was\neventually overshadowed by \nthe lyrics of William ...\n-\nWhat major epic\npoem did Edmund\nSpenser write during \nQueen Elizabeth's\nreign?\nEdmund Spenser composed\n'The\nFaerie\nQueene',\nan \nallegorical epic poem that\nbecame one of the most\nsignificant works of Eliz-\nabethan literature though\nlater\neclipsed\nby\nShake-\nspeare's popularity.\nChristopher\nMarlowe's \n'Hero and Leander' was an-\nother notable Elizabethan\npoem, but unlike Spenser's\nwork\nit \nwasn't\nan\nepic\nallegory.\nWhich poet created\n'Paradise Lost' during\nthe English \nRenaissance?\nJohn Milton authored the\nepic poem 'Paradise Lost'\nin the 17th century, a mon-\numental work that explored\nbiblical\nthemes\nthrough\nblank\nverse\nand\nbecame \na\ncornerstone\nof\nEnglish\nliterature.\nWilliam Blake's 'The Mar-\nriage of Heaven and Hell'\nalso\ndealt\nwith\nreligious\nthemes, though it was more\nprophetic than epic in style \ncompared to Milton's mas-\nterpiece.\n26'),

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page_content='QZhou-Embedding Technical Report\nKingsoft AI\nTable 9: Hard-Negative Generation Example\nquery\npos\nneg\nWhat territory was\nKing Hussein afraid\nIsrael would obtain?\n...Hussein was nonetheless\nwary\nthat\nan\nEgyptian-\nIsraeli war would risk the \nWest Bank's occupation by\nIsrael...\n-\nWhat territory was\nKing Hussein afraid\nIsrael would obtain?\n...Hussein was nonetheless\nwary\nthat\nan\nEgyptian-\nIsraeli war would risk the\nWest Bank's occupation by\nIsrael...\nKing\nHussein\nexpressed\nconcerns\nabout \npotential\nIsraeli\nexpansion\nduring\nthe\nArab-Israeli\nconflicts,\nthough\nhis\nwarnings \nto\nNasser\nwere delayed\nand\ninitially\ndismissed,\nwhile\nother Arab leaders focused \nmore\non\ndirect\nmilitary\npreparations against Israel.\nWhat territory was\nKing Hussein afraid\nIsrael would obtain?\n...Hussein was nonetheless\nwary\nthat\nan\nEgyptian-\nIsraeli war would risk the\nWest Bank's occupation by\nIsrael...\nKing\nHussein\nexpressed \nconcerns\nabout\npotential\nIsraeli territorial expansion\nduring the 1967 tensions,\nthough his warnings were\ndelayed in reaching Nasser\nand\nmixed\nwith\nbroader\nregional

\ntensions,\nwhile\nEgyptian\nmilitary\nmove-\nments in Sinai were already\nunderway\nunder \nAmer's\norders.\n27'),

Document(metadata={'producer': 'pdfTeX-1.40.17', 'creator': 'LaTeX with hyperref package', 'creationdate': '2019-04-17T00:45:22+00:00', 'source': '..\\data\\pdf\\objectdetection.pdf', 'file_path': '..\\data\\pdf\\objectdetection.pdf', 'total_pages': 21, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2019-04-17T00:45:22+00:00', 'trapped': '', 'modDate': 'D:20190417004522Z', 'creationDate': 'D:20190417004522Z', 'page': 0}, page_content='THIS PAPER HAS BEEN ACCEPTED BY IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS FOR PUBLICATION\n1\nObject Detection with Deep Learning: A Review\nZhong-Qiu Zhao, Member, IEEE, Peng Zheng,\nShou-tao Xu, and Xindong Wu, Fellow, IEEE\nAbstract—Due to object detection's close relationship with\nvideo analysis and image understanding, it has attracted much\nresearch attention in recent years. Traditional object detection\nmethods are built on handcrafted features and shallow trainable\narchitectures. Their performance easily stagnates by constructing\ncomplex ensembles which combine multiple low-level image \nfeatures with high-level context from object detectors and scene\nclassifiers. With the rapid development in deep learning, more\npowerful tools, which are able to learn semantic, high-level, \ndeeper features, are introduced to address the problems existing\nin traditional architectures. These models behave differently\nin network architecture, training strategy and optimization \nfunction, etc. In this paper, we provide a review on deep\nlearning based object detection frameworks. Our review begins\nwith a brief introduction on the history of deep learning and\nits representative tool, namely Convolutional Neural Network\n(CNN). Then we focus on typical generic object detection\narchitectures along with some modifications and useful tricks\nto improve detection performance further. As distinct specific\ndetection tasks exhibit different characteristics, we also briefly\nsurvey several specific tasks, including salient object detection, \nface detection and pedestrian detection. Experimental analyses\nare also provided to compare various methods and draw some\nmeaningful conclusions. Finally, several promising directions and\ntasks are provided to serve as guidelines for future work in\nboth object detection and relevant neural network based learning\nsystems.\nIndex Terms—deep learning, object detection, neural network\nI. INTRODUCTION\nT\nO gain a complete image understanding, we should not\nonly concentrate on classifying different images, but\nalso try to precisely estimate the concepts and locations of\nobjects contained in each image. This task is referred as object\ndetection [1][S1], which usually consists of different subtasks\nsuch as face detection [2][S2], pedestrian detection [3][S2]\nand skeleton detection [4][S3]. As one of the fundamental\ncomputer vision problems, object detection is able to provide\nvaluable information for semantic understanding of images\nand videos, and is related to many applications, including\nimage classification [5], [6], human behavior analysis [7][S4],\nface recognition [8][S5] and autonomous driving [9], [10].\nMeanwhile, Inheriting from neural networks and related learn-\ning systems, the progress in these fields will develop neural\nnetwork algorithms, and will also have great impacts on object\ndetection techniques which can be considered as learning\nsystems. [11]–[14][S6]. However, due to large variations in\nviewpoints, poses, occlusions and lighting conditions, it's diffi-\ncult to perfectly accomplish object detection with an additional\nZhong-Qiu Zhao, Peng Zheng and Shou-Tao Xu are with the College of \nComputer Science and Information Engineering, Hefei University of Technol-\nogy, China. Xindong Wu is with the School of Computing and Informatics,\nUniversity of Louisiana at Lafayette, USA.\nManuscript received August xx, 2017; revised xx xx, 2017.\nobject localization task. So much attention has been attracted\nto this field in recent years [15]–[18].\nThe problem definition of object detection is to determine\nwhere objects are located in a given image (object localization)\nand which category each object belongs to (object classifica-\ntion). So the pipeline of traditional object detection models\ncan be mainly divided into three stages: informative region\nselection, feature extraction and classification.\nInformative region selection. As different objects may appear\nin any positions of the image and have different aspect ratios\nor sizes, it is a natural choice to scan the whole image with a\nmulti-scale sliding window. Although this exhaustive strategy\ncan find out all possible positions of the objects, its short-\ncomings are also obvious. Due to a large number of candidate\nwindows, it is computationally expensive and produces too\nmany redundant windows. However, if only a fixed

number of\nsliding window templates are applied, unsatisfactory regions\nmay be produced. \nFeature extraction. To recognize different objects, we need\nto extract visual features which can provide a semantic and\nrobust representation. SIFT [19], HOG [20] and Haar-like [21]\nfeatures are the representative ones. This is due to the fact\nthat these features can produce representations associated with\ncomplex cells in human brain [19]. However, due to the diver-\nsity of appearances, illumination conditions and backgrounds,\nit's difficult to manually design a robust feature descriptor to\nperfectly describe all kinds of objects. \nClassification. Besides, a classifier is needed to distinguish\na target object from all the other categories and to make the\nrepresentations more hierarchical, semantic and informative\nfor visual recognition. Usually, the Supported Vector Machine\n(SVM) [22], AdaBoost [23] and Deformable Part-based Model\n(DPM) [24] are good choices. Among these classifiers, the \nDPM is a flexible model by combining object parts with\ndeformation cost to handle severe deformations. In DPM, with\nthe aid of a graphical model, carefully designed low-level\nfeatures and kinematically inspired part decompositions are\ncombined. And discriminative learning of graphical models\nallows for building high-precision part-based models for a\nvariety of object classes.\nBased on these discriminant local feature descriptors and\nshallow learnable architectures, state of the art results have\nbeen obtained on PASCAL VOC object detection competition\n[25] and real-time embedded systems have been obtained with\na low burden on hardware. However, small gains are obtained\nduring 2010-2012 by only building ensemble systems and\nemploying minor variants of successful methods [15]. This fact\nis due to the following reasons: 1) The generation of candidate\nbounding boxes with a sliding window strategy is redundant,\ninefficient and inaccurate. 2) The semantic gap cannot be \narXiv:1807.05511v2 [cs.CV] 16 Apr 2019'),

Document(metadata={'producer': 'pdfTeX-1.40.17', 'creator': 'LaTeX with hyperref package', 'creationdate': '2019-04-17T00:45:22+00:00', 'source': '..\\data\\pdf\\objectdetection.pdf', 'file_path': '..\\data\\pdf\\objectdetection.pdf', 'total_pages': 21, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2019-04-17T00:45:22+00:00', 'trapped': '', 'modDate': 'D:20190417004522Z', 'creationDate': 'D:20190417004522Z', 'page': 1}, page_content='THIS PAPER HAS BEEN ACCEPTED BY IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS FOR PUBLICATION\n2\nPedestrian \ndetection\nSalient object \ndetection \nFace \ndetection \nGeneric object \ndetection\nObject \ndetection\nBounding box \nregression \nLocal contrast \nSegmentation\nMulti-feature\nBoosting forest\nMulti-scale\nadaption\nFig. 1. The application domains of object detection.\nbridged by the combination of manually engineered low-level\ndescriptors and discriminatively-trained shallow models.\nThanks to the emergency of Deep Neural Networks (DNNs)\n[6][S7], a more significant gain is obtained with the introduc-\ntion of Regions with CNN features (R-CNN) [15]. DNNs, or\nthe most representative CNNs, act in a quite different way from\ntraditional approaches. They have deeper architectures with the\ncapacity to learn more complex features than the shallow ones.\nAlso the expressivity and robust training algorithms allow to\nlearn informative object representations without the need to\ndesign features manually [26].\nSince the proposal of R-CNN, a great deal of improved\nmodels have been suggested, including Fast R-CNN which\njointly optimizes classification and bounding box regression\ntasks [16], Faster R-CNN which takes an additional sub-\nnetwork to generate region proposals [18] and YOLO which\naccomplishes object detection via a fixed-grid regression [17].\nAll of them bring different degrees of detection performance\nimprovements over the primary R-CNN and make real-time\nand accurate object detection become more achievable.\nIn this paper, a systematic review is provided to summarise \nrepresentative models and their different characteristics in\nseveral application domains, including generic object detec-\ntion [15], [16], [18], salient object detection [27], [28], face \ndetection [29]–[31] and pedestrian detection [32], [33]. Their\nrelationships are depicted in Figure 1. Based on basic CNN ar-\nchitectures, generic object detection is achieved with bounding\nbox regression, while salient object detection is accomplished\nwith local contrast enhancement and pixel-level segmentation.\nFace detection and pedestrian detection are closely related\nto generic object detection and mainly accomplished with\nmulti-scale adaption and multi-feature fusion/boosting forest,\nrespectively. The dotted lines indicate that the corresponding\ndomains are associated with each other under certain con-\nditions. It should

be noticed that the covered domains are\ndiversified. Pedestrian and face images have regular structures,\nwhile general objects and scene images have more complex\nvariations in geometric structures and layouts. Therefore,\ndifferent deep models are required by various images.\nThere has been a relevant pioneer effort [34] which mainly\nfocuses on relevant software tools to implement deep learning\ntechniques for image classification and object detection, but\npays little attention on detailing specific algorithms. Different\nfrom it, our work not only reviews deep learning based object\ndetection models and algorithms covering different applica-\ntion domains in detail, but also provides their corresponding\nexperimental comparisons and meaningful analyses.\nThe rest of this paper is organized as follows. In Section\n2, a brief introduction on the history of deep learning and the\nbasic architecture of CNN is provided. Generic object detec-\ntion architectures are presented in Section 3. Then reviews\nof CNN applied in several specific tasks, including salient\nobject detection, face detection and pedestrian detection, are\nexhibited in Section 4-6, respectively. Several promising future\ndirections are proposed in Section 7. At last, some concluding\nremarks are presented in Section 8.\nII. A BRIEF OVERVIEW OF DEEP LEARNING\nPrior to overview on deep learning based object detection\napproaches, we provide a review on the history of deep\nlearning along with an introduction on the basic architecture\nand advantages of CNN.\nA. The History: Birth, Decline and Prosperity\nDeep models can be referred to as neural networks with\ndeep structures. The history of neural networks can date back\nto 1940s [35], and the original intention was to simulate the\nhuman brain system to solve general learning problems in a \nprincipled way. It was popular in 1980s and 1990s with the\nproposal of back-propagation algorithm by Hinton et al. [36].\nHowever, due to the overfitting of training, lack of large scale \ntraining data, limited computation power and insignificance\nin performance compared with other machine learning tools,\nneural networks fell out of fashion in early 2000s.\nDeep learning has become popular since 2006 [37][S7] with\na break through in speech recognition [38]. The recovery of\ndeep learning can be attributed to the following factors.\n• The emergence of large scale annotated training data, such\nas ImageNet [39], to fully exhibit its very large learning \ncapacity;\n• Fast development of high performance parallel computing\nsystems, such as GPU clusters;\n• Significant advances in the design of network structures\nand training strategies. With unsupervised and layerwise\npre-training guided by Auto-Encoder (AE) [40] or Re-\nstricted Boltzmann Machine (RBM) [41], a good initializa-\ntion is provided. With dropout and data augmentation, the\noverfitting problem in training has been relieved [6], [42].\nWith batch normalization (BN), the training of very deep\nneural networks becomes quite efficient [43]. Meanwhile,\nvarious network structures, such as AlexNet [6], Overfeat\n[44], GoogLeNet [45], VGG [46] and ResNet [47], have\nbeen extensively studied to improve the performance. \nWhat prompts deep learning to have a huge impact on the\nentire academic community? It may owe to the contribution of\nHinton's group, whose continuous efforts have demonstrated \nthat deep learning would bring a revolutionary breakthrough\non grand challenges rather than just obvious improvements on\nsmall datasets. Their success results from training a large CNN \non 1.2 million labeled images together with a few techniques\n[6] (e.g., ReLU operation [48] and 'dropout' regularization).\nB. Architecture and Advantages of CNN\nCNN is the most representative model of deep learning [26].\nA typical CNN architecture, which is referred to as VGG16,'),

Document(metadata={'producer': 'pdfTeX-1.40.17', 'creator': 'LaTeX with hyperref package', 'creationdate': '2019-04-17T00:45:22+00:00', 'source': '..\\data\\pdf\\objectdetection.pdf', 'file_path': '..\\data\\pdf\\objectdetection.pdf', 'total_pages': 21, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2019-04-17T00:45:22+00:00', 'trapped': '', 'modDate': 'D:20190417004522Z', 'creationDate': 'D:20190417004522Z', 'page': 2}, page_content='THIS PAPER HAS BEEN ACCEPTED BY IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS FOR PUBLICATION\n3\ncan be found in Fig. S1. Each layer of CNN is known as a \nfeature map. The feature map of the input layer is a 3D matrix\nof pixel intensities for different color channels (e.g. RGB). The\nfeature map of any internal layer is an induced multi-channel \nimage, whose 'pixel' can be viewed as a specific feature. Every\nneuron is connected with a small portion of adjacent neurons\nfrom the previous layer (receptive field). Different types of \ntransformations [6], [49], [50] can be conducted on feature\nmaps, such as filtering and

pooling. Filtering (convolution)\noperation convolutes a filter matrix (learned weights) with\nthe values of a receptive field of neurons and takes a non-\nlinear function (such as sigmoid [51], ReLU) to obtain final\nresponses. Pooling operation, such as max pooling, average\npooling, L2-pooling and local contrast normalization [52],\nsummaries the responses of a receptive field into one value\nto produce more robust feature descriptions.\nWith an interleave between convolution and pooling, an\ninitial feature hierarchy is constructed, which can be fine-tuned\nin a supervised manner by adding several fully connected (FC)\nlayers to adapt to different visual tasks. According to the tasks\ninvolved, the final layer with different activation functions [6]\nis added to get a specific conditional probability for each\noutput neuron. And the whole network can be optimized on\nan objective function (e.g. mean squared error or cross-entropy\nloss) via the stochastic gradient descent (SGD) method. The\ntypical VGG16 has totally 13 convolutional (conv) layers, 3\nfully connected layers, 3 max-pooling layers and a softmax\nclassification layer. The conv feature maps are produced by\nconvoluting 3*3 filter windows, and feature map resolutions\nare reduced with 2 stride max-pooling layers. An arbitrary test\nimage of the same size as training samples can be processed\nwith the trained network. Re-scaling or cropping operations\nmay be needed if different sizes are provided [6].\nThe advantages of CNN against traditional methods can be\nsummarised as follows.\n• Hierarchical feature representation, which is the multi-\nlevel representations from pixel to high-level semantic fea-\ntures learned by a hierarchical multi-stage structure [15],\n[53], can be learned from data automatically and hidden\nfactors of input data can be disentangled through multi-level\nnonlinear mappings.\n• Compared with traditional shallow models, a deeper\narchitecture provides an exponentially increased expressive\ncapability.\n• The architecture of CNN provides an opportunity to\njointly optimize several related tasks together (e.g. Fast R-\nCNN combines classification and bounding box regression\ninto a multi-task leaning manner).\n• Benefitting from the large learning capacity of deep\nCNNs, some classical computer vision challenges can be\nrecast as high-dimensional data transform problems and\nsolved from a different viewpoint.\nDue to these advantages, CNN has been widely applied\ninto many research fields, such as image super-resolution\nreconstruction [54], [55], image classification [5], [56], im-\nage retrieval [57], [58], face recognition [8][S5], pedestrian\ndetection [59]–[61] and video analysis [62], [63].\nIII. GENERIC OBJECT DETECTION\nGeneric object detection aims at locating and classifying \nexisting objects in any one image, and labeling them with\nrectangular bounding boxes to show the confidences of exis-\ntence. The frameworks of generic object detection methods \ncan mainly be categorized into two types (see Figure 2).\nOne follows traditional object detection pipeline, generating\nregion proposals at first and then classifying each proposal into \ndifferent object categories. The other regards object detection\nas a regression or classification problem, adopting a unified\nframework to achieve final results (categories and locations)\ndirectly. The region proposal based methods mainly include\nR-CNN [15], SPP-net [64], Fast R-CNN [16], Faster R-CNN\n[18], R-FCN [65], FPN [66] and Mask R-CNN [67], some of \nwhich are correlated with each other (e.g. SPP-net modifies R-\nCNN with a SPP layer). The regression/classification based\nmethods mainly includes MultiBox [68], AttentionNet [69],\nG-CNN [70], YOLO [17], SSD [71], YOLOv2 [72], DSSD\n[73] and DSOD [74]. The correlations between these two\npipelines are bridged by the anchors introduced in Faster R-\nCNN. Details of these methods are as follows.\nA. Region Proposal Based Framework\nThe region proposal based framework, a two-step process,\nmatches the attentional mechanism of human brain to some \nextent, which gives a coarse scan of the whole scenario firstly\nand then focuses on regions of interest. Among the pre-related\nworks [44], [75], [76], the most representative one is Overfeat \n[44]. This model inserts CNN into sliding window method,\nwhich predicts bounding boxes directly from locations of\nthe topmost feature map after obtaining the confidences of \nunderlying object categories.\n1) R-CNN: It is of significance to improve the quality of \ncandidate bounding boxes and to take a deep architecture to\nextract high-level features. To solve these problems, R-CNN\n[15] was proposed by Ross Girshick in 2014 and obtained a \nmean average precision (mAP) of 53.3% with more than 30%\nimprovement over the previous best result (DPM HSC [77]) on\nPASCAL VOC 2012. Figure 3 shows the flowchart of R-CNN, \nwhich can be divided into three stages as follows.\nRegion proposal generation. The R-CNN adopts selective\nsearch [78] to generate about 2k region proposals for each\nimage. The selective search method relies on simple bottom-up\ngrouping and saliency cues to provide

more accurate candidate\nboxes of arbitrary sizes quickly and to reduce the searching\nspace in object detection [24], [39].\nCNN based deep feature extraction. In this stage, each\nregion proposal is warped or cropped into a fixed resolution\nand the CNN module in [6] is utilized to extract a 4096-\ndimensional feature as the final representation. Due to large\nlearning capacity, dominant expressive power and hierarchical\nstructure of CNNs, a high-level, semantic and robust feature\nrepresentation for each region proposal can be obtained.\nClassification and localization. With pre-trained category-\nspecific linear SVMs for multiple classes, different region pro-\nposals are scored on a set of positive regions and background\n(negative) regions. The scored regions are then adjusted with'),

Document(metadata={'producer': 'pdfTeX-1.40.17', 'creator': 'LaTeX with hyperref package', 'creationdate': '2019-04-17T00:45:22+00:00', 'source': '..\\data\\pdf\\objectdetection.pdf', 'file_path': '..\\data\\pdf\\objectdetection.pdf', 'total_pages': 21, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2019-04-17T00:45:22+00:00', 'trapped': '', 'modDate': 'D:20190417004522Z', 'creationDate': 'D:20190417004522Z', 'page': 3}, page_content='THIS PAPER HAS BEEN ACCEPTED BY IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS FOR PUBLICATION\n4\nGeneric object \ndetection\nRegion proposal \nbased \nRegression/\nClassification \nbased \nR-CNN\n(2014)\nSPP-net\n(2015)\nFRCN \n(2015)\nFaster \nR-CNN\n(2015)\nR-FCN\n(2016)\nFPN\n(2017)\nMask R-CNN \n(2017)\nMultiBox\n(2014)\nAttentionNet\n(2015)\nG-CNN\n(2016)\nYOLO\n(2016)\nSSD \n(2016)\nYOLOv2\n(2017)\nSPP \nlayer\nMulti-\ntask\nRPN\nFCN\nFeature\npyramid \nInstance\nSegmentation\nRegion\nproposal\nUnified\nloss\nDirection\niteration\nJoint Grid \nregression\nRPN\nBN\nMulti-scale\nGrid\nregression\nDSSD\n(2017)\nDSOD \n(2017)\nStem block\nDense block\nResNet101 \nDeconv layers\nFig. 2. Two types of frameworks: region proposal based and regression/classification based. SPP: Spatial Pyramid Pooling [64], FRCN: Faster R-CNN [16],\nRPN: Region Proposal Network [18], FCN: Fully Convolutional Network [65], BN: Batch Normalization [43], Deconv layers: Deconvolution layers [54]\n.\n1. Input \nimage\n2. Extract region \nproposals (~2k)\n3. Compute \nCNN features \naeroplane? no.\n...\nperson? yes.\ntvmonitor? no.\n4. Classify \nregions\nwarped region\n... \nCNN\nR-CNN: Regions with CNN features\nFig. 3. The flowchart of R-CNN [15], which consists of 3 stages: (1) extracts\nbottom-up region proposals, (2) computes features for each proposal using a\nCNN, and then (3) classifies each region with class-specific linear SVMs.\nbounding box regression and filtered with a greedy non-\nmaximum suppression (NMS) to produce final bounding boxes\nfor preserved object locations.\nWhen there are scarce or insufficient labeled data, pre-\ntraining is usually conducted. Instead of unsupervised pre-\ntraining [79], R-CNN firstly conducts supervised pre-training\non ILSVRC, a very large auxiliary dataset, and then takes a\ndomain-specific fine-tuning. This scheme has been adopted by\nmost of subsequent approaches [16], [18].\nIn spite of its improvements over traditional methods and\nsignificance in bringing CNN into practical object detection,\nthere are still some disadvantages.\n• Due to the existence of FC layers, the CNN requires a\nfixed-size (e.g., 227×227) input image, which directly leads\nto the re-computation of the whole CNN for each evaluated\nregion, taking a great deal of time in the testing period.\n• Training of R-CNN is a multi-stage pipeline. At first,\na convolutional network (ConvNet) on object proposals is\nfine-tuned. Then the softmax classifier learned by fine-\ntuning is replaced by SVMs to fit in with ConvNet features.\nFinally, bounding-box regressors are trained.\n• Training is expensive in space and time. Features are\nextracted from different region proposals and stored on the\ndisk. It will take a long time to process a relatively small\ntraining set with very deep networks, such as VGG16. At the\nsame time, the storage memory required by these features\nshould also be a matter of concern.\n• Although selective search can generate region proposals\nwith relatively high recalls, the obtained region proposals\nare still redundant and this procedure is time-consuming\n(around 2 seconds to extract 2k region proposals).\nTo solve these problems, many methods have been pro-\nposed. GOP [80] takes a much faster geodesic based segmen-\ntation to replace traditional graph cuts. MCG [81] searches\ndifferent scales of the image for multiple hierarchical segmen-\ntations and combinatorially groups different regions to produce\nproposals. Instead of extracting visually distinct segments,\nthe edge boxes method [82] adopts the idea that objects are\nmore likely to exist in bounding boxes with fewer contours\nstraggling their boundaries. Also some researches

tried to\nre-rank or refine pre-extracted region proposals to remove\nunnecessary ones and obtained a limited number of valuable\nones, such as DeepBox [83] and SharpMask [84].\nIn addition, there are some improvements to solve the\nproblem of inaccurate localization. Zhang et al. [85] utilized\na bayesian optimization based search algorithm to guide\nthe regressions of different bounding boxes sequentially, and\ntrained class-specific CNN classifiers with a structured loss\nto penalize the localization inaccuracy explicitly. Saurabh\nGupta et al. improved object detection for RGB-D images\nwith semantically rich image and depth features [86], and\nlearned a new geocentric embedding for depth images to\nencode each pixel. The combination of object detectors and\nsuperpixel classification framework gains a promising result\non semantic scene segmentation task. Ouyang et al. proposed\na deformable deep CNN (DeepID-Net) [87] which introduces\na novel deformation constrained pooling (def-pooling) layer \nto impose geometric penalty on the deformation of various\nobject parts and makes an ensemble of models with different\nsettings. Lenc et al. [88] provided an analysis on the role\nof proposal generation in CNN-based detectors and tried to\nreplace this stage with a constant and trivial region generation\nscheme. The goal is achieved by biasing sampling to match\nthe statistics of the ground truth bounding boxes with K-means\nclustering. However, more candidate boxes are required to\nachieve comparable results to those of R-CNN.\n2) SPP-net: FC layers must take a fixed-size input. That’s\nwhy R-CNN chooses to warp or crop each region proposal\ninto the same size. However, the object may exist partly in\nthe cropped region and unwanted geometric distortion may be\nproduced due to the warping operation. These content losses or\ndistortions will reduce recognition accuracy, especially when\nthe scales of objects vary.\nTo solve this problem, He et al. took the theory of spatial\npyramid matching (SPM) [89], [90] into consideration and\nproposed a novel CNN architecture named SPP-net [64]. SPM \ntakes several finer to coarser scales to partition the image into\na number of divisions and aggregates quantized local features\ninto mid-level representations.\nThe architecture of SPP-net for object detection can be\nfound in Figure 4. Different from R-CNN, SPP-net reuses \nfeature maps of the 5-th conv layer (conv5) to project region'),

Document(metadata={'producer': 'pdfTeX-1.40.17', 'creator': 'LaTeX with hyperref package', 'creationdate': '2019-04-17T00:45:22+00:00', 'source': '..\\data\\pdf\\objectdetection.pdf', 'file_path': '..\\data\\pdf\\objectdetection.pdf', 'total_pages': 21, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2019-04-17T00:45:22+00:00', 'trapped': '', 'modDate': 'D:20190417004522Z', 'creationDate': 'D:20190417004522Z', 'page': 4}, page_content='THIS PAPER HAS BEEN ACCEPTED BY IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS FOR PUBLICATION\n5\nspatial pyramid \npooling layer\nfeature maps of conv5\nconvolutional layers\nfixed-length representation\ninput image\nwindow\n……\nfully-connected layers (fc6, fc7)\nFig. 4. The architecture of SPP-net for object detection [64].\nDeep \nConvNet\nConv\nfeature map\nRoI\nprojection\nRoI\npooling\nlayer\nFCs\nRoI feature \nvector\nsoftmax\nbbox\nregressor\nOutputs:\nFC\nFC\nFor each RoI\nFig. 5. The architecture of Fast R-CNN [16].\nproposals of arbitrary sizes to fixed-length feature vectors. The \nfeasibility of the reusability of these feature maps is due to\nthe fact that the feature maps not only involve the strength of\nlocal responses, but also have relationships with their spatial \npositions [64]. The layer after the final conv layer is referred\nto as spatial pyramid pooling layer (SPP layer). If the number\nof feature maps in conv5 is 256, taking a 3-level pyramid,\nthe final feature vector for each region proposal obtained after\nSPP layer has a dimension of 256 × (12 + 22 + 42) = 5376.\nSPP-net not only gains better results with correct estimation\nof different region proposals in their corresponding scales, but\nalso improves detection efficiency in testing period with the\nsharing of computation cost before SPP layer among different \nproposals.\n3) Fast R-CNN: Although SPP-net has achieved impressive\nimprovements in both accuracy and efficiency over R-CNN,\nit still has some notable drawbacks. SPP-net takes almost\nthe same multi-stage pipeline as R-CNN, including feature\nextraction, network fine-tuning, SVM training and bounding-\nbox regressor fitting. So an additional expense on storage space\nis still required. Additionally, the conv layers preceding the\nSPP layer cannot be updated with the fine-tuning algorithm\nintroduced in [64]. As a result, an accuracy drop of very deep \nnetworks is unsurprising. To this end, Girshick [16] introduced\na multi-task loss on classification and bounding box regression\nand proposed a novel CNN architecture named Fast

R-CNN.\nThe architecture of Fast R-CNN is exhibited in Figure 5.\nSimilar to SPP-net, the whole image is processed with conv\nlayers to produce feature maps. Then, a fixed-length feature \nvector is extracted from each region proposal with a region of\ninterest (RoI) pooling layer. The RoI pooling layer is a special\ncase of the SPP layer, which has only one pyramid level. Each \nfeature vector is then fed into a sequence of FC layers before\nfinally branching into two sibling output layers. One output\nlayer is responsible for producing softmax probabilities for \nall C + 1 categories (C object classes plus one 'background'\nclass) and the other output layer encodes refined bounding-\nbox positions with four real-valued numbers. All parameters\nin these procedures (except the generation of region proposals)\nare optimized via a multi-task loss in an end-to-end way.\nThe multi-tasks loss L is defined as below to jointly train \nclassification and bounding-box regression,\nL(p, u, tu, v) = Lcls(p, u) + λ[u ≥1]Lloc(tu, v)\n(1)\nwhere Lcls(p, u) = −log pu calculates the log loss for ground\ntruth class u and pu is driven from the discrete probability\ndistribution p = (p0, · · · , pC) over the C +1 outputs from the \nlast FC layer. Lloc(tu, v) is defined over the predicted offsets\ntu = (tu\nx, tu\ny, tu\nw, tu\nh) and ground-truth bounding-box regression\ntargets v = (vx, vy, vw, vh), where x, y, w, h denote the two\ncoordinates of the box center, width, and height, respectively.\nEach tu adopts the parameter settings in [15] to specify an\nobject proposal with a log-space height/width shift and scale-\ninvariant translation. The Iverson bracket indicator function\n[u ≥1] is employed to omit all background RoIs. To provide\nmore robustness against outliers and eliminate the sensitivity \nin exploding gradients, a smooth L1 loss is adopted to fit\nbounding-box regressors as below \nLloc(tu, v) =\nX\ni x,y,w,h\nsmoothL1(tu\ni −vi)\n(2)\nwhere\nsmoothL1(x) =\n(\n0.5x2\nif |x| < 1\n|x| −0.5\notherwise\n(3)\nTo accelerate the pipeline of Fast R-CNN, another two tricks\nare of necessity. On one hand, if training samples (i.e. RoIs)\ncome from different images, back-propagation through the\nSPP layer becomes highly inefficient. Fast R-CNN samples\nmini-batches hierarchically, namely N images sampled ran-\ndomly at first and then R/N RoIs sampled in each image,\nwhere R represents the number of RoIs. Critically, computa-\ntion and memory are shared by RoIs from the same image in\nthe forward and backward pass. On the other hand, much time\nis spent in computing the FC layers during the forward pass\n[16]. The truncated Singular Value Decomposition (SVD) [91]\ncan be utilized to compress large FC layers and to accelerate\nthe testing procedure.\nIn the Fast R-CNN, regardless of region proposal genera-\ntion, the training of all network layers can be processed in\na single-stage with a multi-task loss. It saves the additional\nexpense on storage space, and improves both accuracy and \nefficiency with more reasonable training schemes.\n4) Faster R-CNN: Despite the attempt to generate candi-\ndate boxes with biased sampling [88], state-of-the-art object\ndetection networks mainly rely on additional methods, such as\nselective search and Edgebox, to generate a candidate pool of\nisolated region proposals. Region proposal computation is also\na bottleneck in improving efficiency. To solve this problem,\nRen et al. introduced an additional Region Proposal Network\n(RPN) [18], [92], which acts in a nearly cost-free way by\nsharing full-image conv features with detection network.\nRPN is achieved with a fully-convolutional network, which\nhas the ability to predict object bounds and scores at each\nposition simultaneously. Similar to [78], RPN takes an image\nof arbitrary size to generate a set of rectangular object propos-\nals. RPN operates on a specific conv layer with the preceding \nlayers shared with object detection network.'),

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obtained in each sliding window and fed into two\nsibling FC layers, namely box-classification layer (cls) and\nbox-regression layer (reg). This architecture is implemented\nwith an n × n conv layer followed by two sibling 1 × 1 conv\nlayers. To increase non-linearity, ReLU is applied to the output\nof the n × n conv layer.\nThe regressions towards true bounding boxes are achieved\nby comparing proposals relative to reference boxes (anchors).\nIn the Faster R-CNN, anchors of 3 scales and 3 aspect ratios\nare adopted. The loss function is similar to (1).\nL(pi, ti) =\n1\nNcls \nX\ni\nLcls(pi, p∗\ni) + λ\n1\nNreg\nX\ni\np∗\ni Lreg(ti, t∗\ni)\n(4)\nwhere pi shows the predicted probability of the i-th anchor\nbeing an object. The ground truth label p∗\ni is 1 if the anchor is\npositive, otherwise 0. ti stores 4 parameterized coordinates of\nthe predicted bounding box while t∗\ni is related to the ground-\ntruth box overlapping with a positive anchor. Lcls is a binary\nlog loss and Lreg is a smoothed L1 loss similar to (2). These\ntwo terms are normalized with the mini-batch size (Ncls)\nand the number of anchor locations (Nreg), respectively. In\nthe form of fully-convolutional networks, Faster R-CNN can\nbe trained end-to-end by back-propagation and SGD in an\nalternate training manner.\nWith the proposal of Faster R-CNN, region proposal based\nCNN architectures for object detection can really be trained\nin an end-to-end way. Also a frame rate of 5 FPS (Frame\nPer Second) on a GPU is achieved with state-of-the-art object\ndetection accuracy on PASCAL VOC 2007 and 2012. How-\never, the alternate training algorithm is very time-consuming\nand RPN produces object-like regions (including backgrounds)\ninstead of object instances and is not skilled in dealing with\nobjects with extreme scales or shapes.\n5) R-FCN: Divided by the RoI pooling layer, a prevalent\nfamily [16], [18] of deep networks for object detection are\ncomposed of two subnetworks: a shared fully convolutional\nsubnetwork (independent of RoIs) and an unshared RoI-wise\nsubnetwork. This decomposition originates from pioneering\nclassification architectures (e.g. AlexNet [6] and VGG16 [46])\nwhich consist of a convolutional subnetwork and several FC\nlayers separated by a specific spatial pooling layer.\nRecent state-of-the-art image classification networks, such\nas Residual Nets (ResNets) [47] and GoogLeNets [45], [93],\nare fully convolutional. To adapt to these architectures, it's\n(a) Featurized image pyramid\npredict\npredict\npredict\npredict\n(b) Single feature map\npredict\n(d) Feature Pyramid Network\npredict\npredict\npredict\n(c) Pyramidal feature hierarchy\npredict\npredict\npredict\nFig. 7. The main concern of FPN [66]. (a) It is slow to use an image pyramid\nto build a feature pyramid. (b) Only single scale features is adopted for faster\ndetection. (c) An alternative to the featurized image pyramid is to reuse the\npyramidal feature hierarchy computed by a ConvNet. (d) FPN integrates both\n(b) and (c). Blue outlines indicate feature maps and thicker outlines denote\nsemantically stronger features.\nnatural to construct a fully convolutional object detection net-\nwork without RoI-wise subnetwork. However, it turns out to be\ninferior with such a naive solution [47]. This inconsistence is\ndue to the dilemma of respecting translation variance in object\ndetection compared with increasing translation invariance in\nimage classification. In other words, shifting an object inside\nan image should be indiscriminative in image classification\nwhile any translation of an object in a bounding box may\nbe meaningful in object detection. A manual insertion of\nthe RoI pooling layer into convolutions can break down\ntranslation invariance at the expense of additional unshared\nregion-wise layers. So Li et al. [65] proposed a region-based \nfully convolutional networks (R-FCN, Fig. S2).\nDifferent from Faster R-CNN, for each category, the last\nconv layer of R-FCN produces a total of k2 position-sensitive\nscore maps with a fixed grid of k × k firstly and a position-\nsensitive RoI pooling layer is then appended to aggregate the \nresponses from these score maps. Finally, in each RoI, k2\nposition-sensitive scores are averaged to produce a C + 1-d\nvector and softmax responses across categories are computed. \nAnother 4k2-d conv layer is appended to obtain class-agnostic\nbounding boxes.\nWith R-FCN, more powerful classification networks can be\nadopted to accomplish object detection in a fully-convolutional\narchitecture by sharing nearly all the layers, and state-of-the-\nart results are obtained on both PASCAL VOC and Microsoft\nCOCO [94] datasets at a test speed of 170ms per image.\n6) FPN: Feature pyramids built upon image pyramids\n(featurized image pyramids) have been widely applied in\nmany object detection systems to improve scale invariance\n[24], [64] (Figure 7(a)). However, training time and memory\nconsumption increase rapidly. To this end, some techniques\ntake only a single input scale to represent high-level semantics\nand increase the robustness to scale changes (Figure 7(b)),\nand image pyramids are built at test time which results in\nan inconsistency between train/test-time inferences [16], [18].\nThe in-

network feature hierarchy in a deep ConvNet produces\nfeature maps of different spatial resolutions while introduces\nlarge semantic gaps caused by different depths (Figure 7(c)).\nTo avoid using low-level features, pioneer works [71], [95]\nusually build the pyramid starting from middle layers or\njust sum transformed feature responses, missing the higher-'),

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detection\nresults. In a word, Mask R-CNN is a flexible and efficient\nframework for instance-level recognition, which can be easily\ngeneralized to other tasks (e.g. human pose estimation [7][S4])\nwith minimal modification.\n8) Multi-task Learning, Multi-scale Representation and \nContextual Modelling:\nAlthough the Faster R-CNN gets\npromising results with several hundred proposals, it still strug-\ngles in small-size object detection and localization, mainly due \nto the coarseness of its feature maps and limited information\nprovided in particular candidate boxes. The phenomenon is\nmore obvious on the Microsoft COCO dataset which consists\nof objects at a broad range of scales, less prototypical images,\nand requires more precise localization. To tackle these prob-\nlems, it is of necessity to accomplish object detection with\nmulti-task learning [100], multi-scale representation [95] and\ncontext modelling [101] to combine complementary informa-\ntion from multiple sources.\nMulti-task Learning learns a useful representation for\nmultiple correlated tasks from the same input [102], [103]. \nBrahmbhatt et al. introduced conv features trained for ob-\nject segmentation and 'stuff' (amorphous categories such as\nground and water) to guide accurate object detection of small\nobjects (StuffNet) [100]. Dai et al. [97] presented Multitask\nNetwork Cascades of three networks, namely class-agnostic\nregion proposal generation, pixel-level instance segmentation \nand regional instance classification. Li et al. incorporated the\nweakly-supervised object segmentation cues and region-based\nobject detection into a multi-stage architecture to fully exploit\nthe learned segmentation features [104].\nMulti-scale Representation combines activations from\nmultiple layers with skip-layer connections to provide seman-\ntic information of different spatial resolutions [66]. Cai et\nal. proposed the MS-CNN [105] to ease the inconsistency\nbetween the sizes of objects and receptive fields with multiple\nscale-independent output layers. Yang et al. investigated two\nstrategies, namely scale-dependent pooling (SDP) and layer-\nwise cascaded rejection classifiers (CRC), to exploit appropri-\nate scale-dependent conv features [33]. Kong et al. proposed\nthe HyperNet to calculate the shared features between RPN\nand object detection network by aggregating and compressing \nhierarchical feature maps from different resolutions into a'),

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\nInstead of concentrating on feature extraction, Ren et al.\nmade a detailed analysis on object classifiers [114], and\nfound that it is of particular importance for object detection\nto construct a deep and convolutional per-region classifier\ncarefully, especially for ResNets [47] and GoogLeNets [45].\nTraditional CNN framework for object detection is not\nskilled in handling significant scale variation, occlusion or\ntruncation, especially when only 2D object detection is in-\nvolved. To address this problem, Xiang et al. proposed a\nnovel subcategory-aware region proposal network [60], which\nguides the generation of region proposals with subcategory \ninformation related to object poses and jointly optimize object\ndetection and subcategory classification.\nOuyang et al. found that the samples from different classes\nfollow a longtailed distribution [115], which indicates that dif-\nferent classes with distinct numbers of samples have different\ndegrees of impacts on feature learning. To this end, objects are\nfirstly clustered into visually similar class groups, and then a\nhierarchical feature learning scheme is adopted to learn deep\nrepresentations for each group separately.\nIn order to minimize computational cost and achieve the\nstate-of-the-art performance, with the 'deep and thin' design\nprinciple and following the pipeline of Fast R-CNN, Hong et\nal. proposed the architecture of PVANET [116], which adopts\nsome building blocks including concatenated ReLU [117],\nInception [45], and HyperNet [101] to reduce the expense on\nmulti-scale feature extraction and trains the network with batch\nnormalization [43], residual connections [47], and learning\nrate scheduling based on plateau detection [47]. The PVANET\nachieves the state-of-the-art performance and can be processed\nin real time on Titan X GPU (21 FPS).\nB. Regression/Classification Based Framework\nRegion proposal based frameworks are composed of sev-\neral correlated stages, including region proposal generation,\nfeature extraction with CNN, classification and bounding box\nregression, which are usually trained separately. Even in recent\nend-to-end module Faster R-CNN, an alternative training is\nstill required to obtain shared convolution parameters between \nRPN and detection network. As a result, the time spent in\nhandling different components becomes the bottleneck in real-\ntime application.\nOne-step\nframeworks\nbased\non\nglobal \nregres-\nsion/classification, mapping straightly from image pixels\nto bounding box coordinates and class probabilities, can\nreduce time expense. We firstly reviews some pioneer CNN\nmodels, and then focus on two significant frameworks,\nnamely You only look once (YOLO) [17] and Single Shot\nMultiBox Detector (SSD) [71].\n1) Pioneer Works: Previous to YOLO and SSD, many\nresearchers have already tried to model object detection as\na regression or classification task.\nSzegedy et al. formulated object detection task as a DNN-\nbased regression [118], generating a binary mask for the\ntest image and extracting detections with a simple bounding\nbox inference. However, the model has difficulty in handling\noverlapping objects, and bounding boxes generated by direct\nupsampling is far from perfect.\nPinheiro et al. proposed a CNN model with two branches:\none generates class agnostic segmentation masks and the\nother predicts the likelihood of a given patch centered on\nan object [119]. Inference is efficient since class scores and\nsegmentation can be obtained in a single model with most of\nthe CNN operations shared.\nErhan et al. proposed regression based MultiBox to produce\nscored class-agnostic region proposals [68], [120]. A unified\nloss was introduced to bias both localization and confidences\nof multiple components to predict the coordinates of class-\nagnostic bounding boxes. However, a large quantity of addi-\ntional parameters are introduced to the final layer.\nYoo et al. adopted an iterative classification approach to\nhandle object detection and proposed an impressive end-to-\nend CNN architecture named AttentionNet [69]. Starting from\nthe top-left (TL) and bottom-right (BR) corner of an image, \nAttentionNet points to a target object by generating quantized\nweak directions and converges to an accurate object bound-\nary box with an ensemble of iterative predictions. However,\nthe model becomes quite inefficient when handling multiple\ncategories with a progressive two-step procedure.\nNajibi et al. proposed a proposal-free iterative grid based \nobject detector (G-CNN), which models object detection as'),

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Fig. 9. Main idea of YOLO [17].\nfinding a path from a fixed grid to boxes tightly surrounding\nthe objects [70]. Starting with a fixed multi-scale bounding box \ngrid, G-CNN trains a regressor to move and scale elements of\nthe grid towards objects iteratively. However, G-CNN has a\ndifficulty in dealing with small or highly overlapping objects. \n2) YOLO: Redmon et al. [17] proposed a novel framework\ncalled YOLO, which makes use of the whole topmost feature\nmap to predict both confidences for multiple categories and \nbounding boxes. The basic idea of YOLO is exhibited in\nFigure 9. YOLO divides the input image into an S × S grid and\neach grid cell is responsible for predicting the object centered\nin that grid cell. Each grid cell predicts B bounding boxes\nand their corresponding confidence scores. Formally, confi-\ndence scores are defined as Pr(Object) ∗ IOUtruth\npred , which \nindicates how likely there exist objects (Pr(Object) ≥0) and\nshows confidences of its prediction (IOU truth\npred). At the same\ntime, regardless of the number of boxes, C conditional class\nprobabilities (Pr(Classi|Object)) should also be predicted in\neach grid cell. It should be noticed that only the contribution\nfrom the grid cell containing an object is calculated.\nAt test time, class-specific confidence scores for each box\nare achieved by multiplying the individual box confidence\npredictions with the conditional class probabilities as follows.\nPr(Object) ∗ IOUtruth\npred ∗ Pr(Classi|Object)\n= Pr(Classi) ∗ IOUtruth\npred \n(5)\nwhere the existing probability of class-specific objects in the\nbox and the fitness between the predicted box and the object\nare both taken into consideration.\nDuring training, the following loss function is optimized,\nλcoord\nS2\nX\ni=0\nB\nX\nj=0\n1obj\nij\n\x02\n(xi −ˆxi)2 + (yi −ˆyi)2\x03\n+λcoord\nS2\nX\ni=0\nB\nX\nj=0\n1obj\nij\n"\x12√wi −\np\nˆwi)2 + (\np\nhi −\nq\nˆhi\n\x132#\n+\nS2\nX\ni=0\nB\nX\nj=0\n1obj\nij\n\x10\nCi −ˆCi\n\x112\n +λnoobj\nS2\nX\ni=0\nB\nX\nj=0\n1noobj\nij\n\x10\nCi −ˆCi\n\x112\n+\nS2\nX\ni=0\n1obj\ni \nX\nc∈ classes\n(pi(c) −ˆpi(c))2\n(6)\nIn a certain cell i, (xi, yi) denote the center of the box relative\nto the bounds of the grid cell, (wi, hi) are the normalized width\nand height relative to the image size, Ci represents confidence\nscores, 1obj\ni\nindicates the existence of objects and 1obj\nij\ndenotes\nthat the prediction is conducted by the jth bounding box\npredictor. Note that only when an object is present in that grid\ncell, the loss function penalizes classification errors. Similarly,\nwhen the predictor is 'responsible' for the ground truth box\n(i.e. the highest IoU of any predictor in that grid cell is\nachieved), bounding box coordinate errors are penalized. \nThe YOLO consists of 24 conv layers and 2 FC layers,\nof which some conv layers construct ensembles of inception\nmodules with 1 × 1 reduction layers followed by 3 × 3 conv\nlayers. The network can process images in real-time at 45\nFPS and a simplified version Fast YOLO can reach 155 FPS\nwith better results than other real-time detectors. Furthermore,\nYOLO produces fewer false positives on background, which\nmakes the cooperation with Fast R-CNN become possible. An\nimproved version, YOLOv2, was later proposed in [72], which\nadopts several impressive strategies, such as BN, anchor boxes,\ndimension cluster and multi-scale training. \n3) SSD: YOLO has a difficulty in dealing with small\nobjects in groups, which is caused by strong spatial constraints\nimposed on bounding box predictions [17]. Meanwhile, YOLO \nstruggles to generalize to objects in new/unusual aspect ratios/\nconfigurations and produces relatively coarse features due to\nmultiple downsampling operations.\nAiming at these problems, Liu et al. proposed a Single Shot\nMultiBox Detector (SSD) [71], which was inspired by the\nanchors adopted in MultiBox [68], RPN [18] and multi-scale\nrepresentation [95]. Given a specific feature map, instead of\nfixed grids adopted in YOLO, the SSD takes advantage of a set \nof default anchor boxes with different aspect ratios and scales\nto discretize the output space of bounding boxes. To handle\nobjects with various sizes, the network fuses predictions from \nmultiple feature maps with different resolutions .\nThe architecture of SSD is demonstrated in Figure 10. Given\nthe VGG16 backbone architecture, SSD adds several feature\nlayers to the end of the network, which are responsible for\npredicting the offsets to default boxes with different scales and\naspect ratios and their associated confidences. The network is\ntrained with a weighted sum of localization loss (e.g. Smooth\nL1) and confidence loss (e.g. Softmax), which is similar to\n(1). Final detection results are obtained by conducting NMS\non multi-scale refined bounding boxes.\nIntegrating with hard negative mining, data augmentation\nand a larger number of carefully chosen default anchors,\nSSD significantly outperforms the Faster R-CNN in terms of\naccuracy on PASCAL VOC and COCO, while being three\ntimes faster. The SSD300

(input image size is 300×300) runs\nat 59 FPS, which is more accurate and efficient than YOLO. \nHowever, SSD is not skilled at dealing with small objects,\nwhich can be relieved by adopting better feature extractor\nbackbone (e.g. ResNet101), adding deconvolution layers with\nskip connections to introduce additional large-scale context\n[73] and designing better network structure (e.g. Stem Block\nand Dense Block) [74].'),

Document(metadata={'producer': 'pdfTeX-1.40.17', 'creator': 'LaTeX with hyperref package', 'creationdate': '2019-04-17T00:45:22+00:00', 'source': '..\\data\\pdf\\objectdetection.pdf', 'file_path': '..\\data\\pdf\\objectdetection.pdf', 'total_pages': 21, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2019-04-17T00:45:22+00:00', 'trapped': '', 'modDate': 'D:20190417004522Z', 'creationDate': 'D:20190417004522Z', 'page': 9}, page_content='THIS PAPER HAS BEEN ACCEPTED BY IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS FOR PUBLICATION\n10\nFig. 10. The architecture of SSD 300 [71]. SSD adds several feature layers to the end of VGG16 backbone network to predict the offsets to default anchor \nboxes and their associated confidences. Final detection results are obtained by conducting NMS on multi-scale refined bounding boxes.\nC. Experimental Evaluation\nWe compare various object detection methods on three\nbenchmark datasets, including PASCAL VOC 2007 [25], \nPASCAL VOC 2012 [121] and Microsoft COCO [94]. The\nevaluated approaches include R-CNN [15], SPP-net [64], Fast\nR-CNN [16], NOC [114], Bayes [85], MR-CNN&S-CNN\n[105], Faster R-CNN [18], HyperNet [101], ION [95], MS-\nGR [104], StuffNet [100], SSD300 [71], SSD512 [71], OHEM\n[113], SDP+CRC [33], GCNN [70], SubCNN [60], GBD-Net\n[109], PVANET [116], YOLO [17], YOLOv2 [72], R-FCN\n[65], FPN [66], Mask R-CNN [67], DSSD [73] and DSOD\n[74]. If no specific instructions for the adopted framework\nare provided, the utilized model is a VGG16 [46] pretrained\non 1000-way ImageNet classification task [39]. Due to the\nlimitation of paper length, we only provide an overview, in-\ncluding proposal, learning method, loss function, programming\nlanguage and platform, of the prominent architectures in Table\nI. Detailed experimental settings, which can be found in the\noriginal papers, are missed. In addition to the comparisons of\ndetection accuracy, another comparison is provided to evaluate\ntheir test consumption on PASCAL VOC 2007.\n1) PASCAL VOC 2007/2012: PASCAL VOC 2007 and \n2012 datasets consist of 20 categories. The evaluation terms\nare Average Precision (AP) in each single category and mean\nAverage Precision (mAP) across all the 20 categories. Com-\nparative results are exhibited in Table II and III, from which\nthe following remarks can be obtained.\n• If incorporated with a proper way, more powerful back-\nbone CNN models can definitely improve object detection\nperformance (the comparison among R-CNN with AlexNet, \nR-CNN with VGG16 and SPP-net with ZF-Net [122]).\n• With the introduction of SPP layer (SPP-net), end-to-\nend multi-task architecture (FRCN) and RPN (Faster R-\nCNN), object detection performance is improved gradually\nand apparently.\n• Due to large quantities of trainable parameters, in order to\nobtain multi-level robust features, data augmentation is very\nimportant for deep learning based models (Faster R-CNN\nwith ‘07’ ,‘07+12’ and ‘07+12+coco’).\n• Apart from basic models, there are still many other factors\naffecting object detection performance, such as multi-scale\nand multi-region feature extraction (e.g. MR-CNN), modi-\nfied classification networks (e.g. NOC), additional informa-\ntion from other correlated tasks (e.g. StuffNet, HyperNet),\nmulti-scale representation (e.g. ION) and mining of hard\nnegative samples (e.g. OHEM).\n• As YOLO is not skilled in producing object localizations\nof high IoU, it obtains a very poor result on VOC 2012.\nHowever, with the complementary information from Fast\nR-CNN (YOLO+FRCN) and the aid of other strategies,\nsuch as anchor boxes, BN and fine grained features, the\nlocalization errors are corrected (YOLOv2).\n• By combining many recent tricks and modelling the whole\nnetwork as a fully convolutional one, R-FCN achieves a\nmore obvious improvement of detection performance over\nother approaches.\n2) Microsoft COCO: Microsoft COCO is composed of\n300,000 fully segmented images, in which each image has \nan average of 7 object instances from a total of 80 categories.\nAs there are a lot of less iconic objects with a broad range\nof scales and a stricter requirement on object localization, \nthis dataset is more challenging than PASCAL 2012. Object\ndetection performance is evaluated by AP computed under\ndifferent degrees of IoUs and on different object sizes. The \nresults are shown in Table IV.\nBesides similar remarks to those of PASCAL VOC, some\nother conclusions can be drawn as follows from Table IV.\n• Multi-scale training and test are beneficial

in improv-\ning object detection performance, which provide additional\ninformation in different resolutions (R-FCN). FPN and\nDSSD provide some better ways to build feature pyramids\nto achieve multi-scale representation. The complementary\ninformation from other related tasks is also helpful for\naccurate object localization (Mask R-CNN with instance\nsegmentation task).\n•\nOverall,\nregion\nproposal\nbased\nmethods,\nsuch\nas\nFaster R-CNN and R-FCN, perform better than regres-\nsion/classfication based approaches, namely YOLO and\nSSD, due to the fact that quite a lot of localization errors\nare produced by regression/classfication based approaches.\n• Context modelling is helpful to locate small objects,\nwhich provides additional information by consulting nearby\nobjects and surroundings (GBD-Net and multi-path).\n• Due to the existence of a large number of nonstandard\nsmall objects, the results on this dataset are much worse\nthan those of VOC 2007/2012. With the introduction of\nother powerful frameworks (e.g. ResNeXt [123]) and useful\nstrategies (e.g. multi-task learning [67], [124]), the perfor-\nmance can be improved.\n• The success of DSOD in training from scratch stresses the \nimportance of network design to release the requirements\nfor perfect pre-trained classifiers on relevant tasks and large\nnumbers of annotated samples.\n3) Timing Analysis: Timing analysis (Table V) is conducted\non Intel i7-6700K CPU with a single core and NVIDIA Titan'),

Document(metadata={'producer': 'pdfTeX-1.40.17', 'creator': 'LaTeX with hyperref package', 'creationdate': '2019-04-17T00:45:22+00:00', 'source': '..\\data\\pdf\\objectdetection.pdf', 'file_path': '..\\data\\pdf\\objectdetection.pdf', 'total_pages': 21, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2019-04-17T00:45:22+00:00', 'trapped': '', 'modDate': 'D:20190417004522Z', 'creationDate': 'D:20190417004522Z', 'page': 10}, page_content='THIS PAPER HAS BEEN ACCEPTED BY IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS FOR PUBLICATION\n11\nTABLE I\nAN OVERVIEW OF PROMINENT GENERIC OBJECT DETECTION ARCHITECTURES.\nFramework\nProposal\nMulti-scale Input\nLearning Method \nLoss Function\nSoftmax Layer\nEnd-to-end Train\nPlatform\nLanguage\nR-CNN [15]\nSelective Search\n-\nSGD,BP\nHinge loss (classification),Bounding box regression\n+\n-\nCaffe\nMatlab\nSPP-net [64]\nEdgeBoxes\n+\nSGD\nHinge loss (classification),Bounding box regression\n+\n-\nCaffe\nMatlab\nFast RCNN [16]\nSelective Search\n+\nSGD\nClass Log loss +bounding box regression\n+\n-\nCaffe\nPython\nFaster R-CNN [18]\nRPN\n+\nSGD\nClass Log loss+bounding box regression\n+\n+\nCaffe\nPython/Matlab\nR-FCN [65]\nRPN\n+\nSGD \nClass Log loss+bounding box regression\n-\n+\nCaffe\nMatlab\nMask R-CNN [67]\nRPN\n+ \nSGD\nClass Log loss+bounding box regression\n+\n+\nTensorFlow/Keras\nPython\n +Semantic sigmoid loss\nFPN [66]\nRPN\n+\nSynchronized SGD\nClass Log loss+bounding box regression\n+\n+\nTensorFlow\nPython\nYOLO [17]\n-\n-\nSGD\nClass sum-squared error loss+bounding box regression\n+\n+\nDarknet\nC\n+object confidence+background confidence \nSSD [71]\n-\n-\nSGD\nClass softmax loss+bounding box regression\n-\n+\nCaffe\nC++ \nYOLOv2 [72]\n-\n-\nSGD\nClass sum-squared error loss+bounding box regression\n+\n+ \nDarknet\nC\n+object confidence+background confidence\n* ‘+’ denotes that corresponding techniques are employed while ‘-’ denotes that this technique is not considered. It should be noticed that R-CNN and SPP-net can not be trained end-to-end with a multi-task loss while the \nother architectures are based on multi-task joint training. As most of these architectures are re-implemented on different platforms with various programming languages, we only list the information associated with the versions\nby the referenced authors.\nTABLE II \nCOMPARATIVE RESULTS ON VOC 2007 TEST SET (%).\nMethods\nTrained on\nareo\nbike \nbird\nboat\nbottle\nbus\ncar\ncat\nchair\ncow\ntable\ndog\nhorse\nmbike\nperson\nplant \nsheep\nsofa\ntrain\ntv\nmAP\nR-CNN (Alex) [15]\n07\n68.1\n72.8\n56.8\n43.0\n36.8\n66.3\n74.2\n67.6\n34.4\n63.5\n54.5\n61.2\n69.1\n 68.6\n58.7\n33.4\n62.9\n51.1\n62.5\n68.6\n58.5\nR-CNN(VGG16) [15]\n07\n73.4\n77.0\n63.4\n45.4\n44.6\n75.1\n78.1\n79.8\n40.5\n73.7\n62.2\n79.4\n78.1\n 73.1\n64.2\n35.6\n66.8\n67.2\n70.4\n71.1\n66.0\nSPP-net(ZF) [64]\n07\n68.5\n71.7\n58.7\n41.9\n42.5\n67.7\n72.1\n73.8\n34.7\n67.0\n63.4\n66.0\n72.5\ n71.3\n58.9\n32.8\n60.9\n56.1\n67.9\n68.8\n60.9\nGCNN [70]\n07\n68.3\n77.3\n68.5\n52.4\n38.6\n78.5\n79.5\n81.0\n47.1\n73.6\n64.5\n77.2\n80.5\n 75.8\n66.6\n34.3\n65.2\n64.4\n75.6\n66.4\n66.8\nBayes [85]\n07\n74.1\n83.2\n67.0\n50.8\n51.6\n76.2\n81.4\n77.2\n48.1\n78.9\n65.6\n77.3\n78.4\n7

5.1\n70.1\n41.4\n69.6\n60.8\n70.2\n73.7\n68.5\nFast R-CNN
[16]\n07+12\n77.0\n78.1\n69.3\n59.4\n38.3\n81.6\n78.6\n86.7\n42.8\n78.8\n68.9\n84.7\n82
.0\n76.6\n69.9\n31.8\n70.1\n74.8\n80.4\n70.4\n70.0\nSDP+CRC
[33]\n07\n76.1\n79.4\n68.2\n52.6\n46.0\n78.4\n78.4\n81.0\n46.7\n73.5\n65.3\n78.6\n81.0\
n76.7\n77.3\n39.0\n65.1\n67.2\n77.5\n70.3\n68.9\nSubCNN
[60]\n07\n70.2\n80.5\n69.5\n60.3\n47.9\n79.0\n78.7\n84.2\n48.5\n73.9\n63.0\n82.7\n80.6\n7
6.0\n70.2\n38.2\n62.4\n67.7\n77.7\n60.5\n68.5\nStuffNet30
[100]\n07\n72.6\n81.7\n70.6\n60.5\n53.0\n81.5\n83.7\n83.9\n52.2\n78.9\n70.7\n85.0\n85.7\
n77.0\n78.7\n42.2\n73.6\n69.2\n79.2\n73.8\n72.7\nNOC
[114]\n07+12\n76.3\n81.4\n74.4\n61.7\n60.8\n84.7\n78.2\n82.9\n53.0\n79.2\n69.2\n83.2\n8
3.2\n78.5\n68.0\n45.0\n71.6\n76.7\n82.2\n75.7\n73.3\nMR-CNN&S-CNN
[110]\n07+12\n80.3\n84.1\n78.5\n70.8\n68.5\n88.0\n85.9\n87.8\n60.3\n85.2\n73.7\n87.2\n86
.5\n85.0\n76.4\n48.5\n76.3\n75.5\n85.0\n81.0\n78.2\nHyperNet
[101]\n07+12\n77.4\n83.3\n75.0\n69.1\n62.4\n83.1\n87.4\n87.4\n57.1\n79.8\n71.4\n85.1\n8
5.1\n80.0\n79.1\n51.2\n79.1\n75.7\n80.9\n76.5\n76.3\nMS-GR
[104]\n07+12\n80.0\n81.0\n77.4\n72.1\n64.3\n88.2\n88.1\n88.4\n64.4\n85.4\n73.1\n87.3\n8
7.4\n85.1\n79.6\n50.1\n78.4\n79.5\n86.9\n75.5\n78.6\nOHEM+Fast R-CNN
[113]\n07+12\n80.6\n85.7\n79.8\n69.9\n60.8\n88.3\n87.9\n89.6\n59.7\n85.1\n76.5\n87.1\n87
.3\n82.4\n78.8\n53.7\n80.5\n78.7\n84.5\n80.7\n78.9\nION [95]\n07+12+S
\n80.2\n85.2\n78.8\n70.9\n62.6\n86.6\n86.9\n89.8\n61.7\n86.9\n76.5\n88.4\n87.5\n83.4\n80.
5\n52.4\n78.1\n77.2\n86.9\n83.5\n79.2\nFaster R-CNN
[18]\n07\n70.0\n80.6\n70.1\n57.3\n49.9\n78.2\n80.4\n82.0\n52.2\n75.3\n67.2\n80.3\n79.8\n7
5.0\n76.3\n39.1\n68.3\n67.3\n81.1\n67.6\n69.9\nFaster R-CNN
[18]\n07+12\n76.5\n79.0\n70.9\n65.5\n52.1\n83.1\n84.7\n86.4\n52.0\n81.9\n65.7\n84.8\n84.
6\n77.5\n76.7\n38.8\n73.6\n73.9\n83.0\n72.6\n73.2\nFaster R-CNN [18]\n07+12+COCO
\n84.3\n82.0\n77.7\n68.9\n65.7\n88.1\n88.4\n88.9\n63.6\n86.3\n70.8\n85.9\n87.6\n80.1\n82
.3\n53.6\n80.4\n75.8\n86.6\n78.9\n78.8\nSSD300 [71]\n07+12+COCO
\n80.9\n86.3\n79.0\n76.2\n57.6\n87.3\n88.2\n88.6\n60.5\n85.4\n76.7\n87.5\n89.2\n84.5\n81
.4\n55.0\n81.9\n81.5\n85.9\n78.9\n79.6\nSSD512 [71]\n07+12+COCO
\n86.6\n88.3\n82.4\n76.0\n66.3\n88.6\n88.9\n89.1\n65.1\n88.4\n73.6\n86.5\n88.9\n85.3\n84
.6\n59.1\n85.0\n80.4\n87.4\n81.2\n81.6\n* ‘07’: VOC2007 trainval, ‘07+12’: union of VOC2007 and VOC2012 trainval, ‘07+12+COCO’: trained on COCO trainval35k at first and then fine-tuned on 07+12. The S in ION ‘07+12+S’ denotes SBD segmentation labels.\nTABLE III\nCOMPARATIVE RESULTS ON VOC 2012 TEST SET (%).\nMethods\nTrained on\nareo\nbike\nbird\nboat\nbottle
\nbus\ncar\ncat\nchair\ncow\ntable\ndog\nhorse\nmbike\nperson\nplant\nsheep\nsofa
\ntrain\ntv\nmAP\nR-CNN(Alex)
[15]\n12\n71.8\n65.8\n52.0\n34.1\n32.6\n59.6\n60.0\n69.8\n27.6\n52.0\n41.7\n69.6\n61.3\n
68.3\n57.8\n29.6\n57.8\n40.9\n59.3\n54.1\n53.3\nR-CNN(VGG16)
[15]\n12\n79.6\n72.7\n61.9\n41.2\n41.9\n65.9\n66.4\n84.6\n38.5\n67.2\n46.7\n82.0\n74.8\n
76.0\n65.2\n35.6\n65.4\n54.2\n67.4\n60.3\n62.4\nBayes
[85]\n12\n82.9\n76.1\n64.1\n44.6\n49.4\n70.3\n71.2\n84.6\n42.7\n68.6\n55.8\n82.7\n77.1\n7
9.9\n68.7\n41.4\n69.0\n60.0\n72.0\n66.2\n66.4\nFast R-CNN [16]\n07+
+12\n82.3\n78.4\n70.8\n52.3\n38.7\n77.8\n71.6\n89.3\n44.2\n73.0\n55.0\n87.5\n80.5\n80.8\
n72.0\n35.1\n68.3\n65.7\n80.4\n64.2\n68.4\nSutffNet30
[100]\n12\n83.0\n76.9\n71.2\n51.6\n50.1\n76.4\n75.7\n87.8\n48.3\n74.8\n55.7\n85.7\n81.2\
n80.3\n79.5\n44.2\n71.8\n61.0\n78.5\n65.4\n70.0\nNOC
[114]\n07+12\n82.8\n79.0\n71.6\n52.3\n53.7\n74.1\n69.0\n84.9\n46.9\n74.3\n53.1\n85.0\n8
1.3\n79.5\n72.2\n38.9\n72.4\n59.5\n76.7\n68.1\n68.8\nMR-CNN&S-CNN [110]\n07+
+12\n85.5\n82.9\n76.6\n57.8\n62.7\n79.4\n77.2\n86.6\n55.0\n79.1\n62.2\n87.0\n83.4\n84.7\
n78.9\n45.3\n73.4\n65.8\n80.3\n74.0\n73.9\nHyperNet [101]\n07+
+12\n84.2\n78.5\n73.6\n55.6\n53.7\n78.7\n79.8\n87.7\n49.6\n74.9\n52.1\n86.0\n81.7\n83.3
\n81.8\n48.6\n73.5\n59.4\n79.9\n65.7\n71.4\nOHEM+Fast R-CNN [113]\n07++12+coco
\n90.1\n87.4\n79.9\n65.8\n66.3\n86.1\n85.0\n92.9\n62.4\n83.4\n69.5\n90.6\n88.9\n88.9\n83
.6\n59.0\n82.0\n74.7\n88.2\n77.3\n80.1\nION [95]\n07+12+S
\n87.5\n84.7\n76.8\n63.8\n58.3\n82.6\n79.0\n90.9\n57.8\n82.0\n64.7\n88.9\n86.5\n84.7\n82.
3\n51.4\n78.2\n69.2\n85.2\n73.5\n76.4\nFaster R-CNN [18]\n07+

+12\n84.9\n79.8\n74.3\n53.9\n49.8\n77.5\n75.9\n88.5\n45.6\n77.1\n55.3\n86.9\n81.7\n80.9\n79.6\n40.1\n72.6\n60.9\n81.2\n61.5\n70.4\nFaster R-CNN [18]\n07++12+coco\n87.4\n83.6\n76.8\n62.9\n59.6\n81.9\n82.0\n91.3\n54.9\n82.6\n59.0\n89.0\n85.5\n84.7\n84.1\n52.2\n78.9\n65.5\n85.4\n70.2\n75.9\nYOLO [17]\n07++12\n77.0\n67.2\n57.7\n38.3\n22.7\n68.3\n55.9\n81.4\n36.2\n60.8\n48.5\n77.2\n72.3\n71.3\n63.5\n28.9\n52.2\n54.8\n73.9\n50.8\n57.9\nYOLO+Fast R-CNN [17]\n07++12\n83.4\n78.5\n73.5\n55.8\n43.4\n79.1\n73.1\n89.4\n49.4\n75.5\n57.0\n87.5\n80.9\n81.0\n74.7\n41.8\n71.5\n68.5\n82.1\n67.2\n70.7\nYOLOv2 [72]\n07++12+coco\n88.8\n87.0\n77.8\n64.9\n51.8\n85.2\n79.3\n93.1\n64.4\n81.4\n70.2\n91.3\n88.1\n87.2\n81.0\n57.7\n78.1\n71.0\n88.5\n76.8\n78.2\nSSD300 [71]\n07++12+coco\n91.0\n86.0\n78.1\n65.0\n55.4\n84.9\n84.0\n93.4\n62.1\n83.6\n67.3\n91.3\n88.9\n88.6\n85.6\n54.7\n83.8\n77.3\n88.3\n76.5\n79.3\nSSD512 [71]\n07++12+coco\n91.4\n88.6\n82.6\n71.4\n63.1\n87.4\n88.1\n93.9\n66.9\n86.6\n66.3\n92.0\n91.7\n90.8\n88.5\n60.9\n87.0\n75.4\n90.2\n80.4\n82.2\nR-FCN (ResNet101) [16]\n07++12+coco\n92.3\n89.9\n86.7\n74.7\n75.2\n86.7\n89.0\n95.8\n70.2\n90.4\n66.5\n95.0\n93.2\n92.1\n91.1\n71.0\n89.7\n76.0\n92.0\n83.4\n85.0\n* ‘07++12’: union of VOC2007 trainval and test and VOC2012 trainval. ‘07++12+COCO’: trained on COCO trainval35k at first then fine-tuned on 07++12.\nTABLE IV\nCOMPARATIVE RESULTS ON MICROSOFT COCO TEST DEV SET (%).\nMethods\nTrained on 0.5:0.95 0.5 0.75\nS\nM\nL\n1\n10\n100\nS\nM\nL\nFast R-CNN [16]\ntrain\n20.5\n39.9 19.4 4.1 20.0 35.8 21.3 29.4 30.1 7.3 32.1 52.0\nION [95]\ntrain\n23.6\n43.2 23.6 6.4 24.1 38.3 23.2 32.7 33.5 10.1 37.7 53.6\nNOC+FRCN(VGG16) [114]\ntrain\n21.2\n41.5 19.7\n-\n-\n-\n-\n-\n-\n-\n-\n-\n-\nNOC+FRCN(Google) [114]\ntrain\n24.8\n44.4 25.2\n-\n-\n-\n-\n-\n-\n-\n-\n-\n-\nNOC+FRCN (ResNet101) [114]\ntrain\n27.2\n48.4 27.6\n-\n-\n-\n-\n-\n-\n-\n-\n-\n-\nGBD-Net [109]\ntrain\n27.0\n45.8\n-\n-\n-\n-\n-\n-\n-\n-\n-\n-\n-\nOHEM+FRCN [113]\ntrain\n22.6\n42.5 22.2 5.0 23.7 34.6\n-\n-\n-\n-\n-\n-\n-\nOHEM+FRCN* [113]\ntrain\n24.4\n44.4 24.8 7.1 26.4 37.9\n-\n-\n-\n-\n-\n-\n-\nOHEM+FRCN* [113]\ntrainval\n25.5\n45.9 26.1 7.4 27.7 38.5\n-\n-\n-\n-\n-\n-\n-\nFaster R-CNN [18]\ntrainval\n24.2\n45.3 23.5 7.7 26.4 37.1 23.8 34.0 34.6 12.0 38.5 54.4\nYOLOv2 [72]\ntrainval35k\n21.6\n44.0 19.2 5.0 22.4 35.5 20.7 31.6 33.3 9.8 36.5 54.4\nSSD300 [71]\ntrainval35k\n23.2\n41.2 23.4 5.3 23.2 39.6 22.5 33.2 35.3 9.6 37.6 56.5\nSSD512 [71]\ntrainval35k\n26.8\n46.5 27.8 9.0 28.9 41.9 24.8 37.5 39.8 14.0 43.5 59.0\nR-FCN (ResNet101) [65]\ntrainval\n29.2\n51.5\n-\n10.8 32.8 45.0\n-\n-\n-\n-\n-\n-\n-\nR-FCN*(ResNet101) [65]\ntrainval\n29.9\n51.9\n-\n10.4 32.4 43.3\n-\n-\n-\n-\n-\n-\n-\nR-FCN**(ResNet101) [65]\ntrainval\n31.5\n53.2\n-\n14.3 35.5 44.2\n-\n-\n-\n-\n-\n-\n-\nMulti-path [112]\ntrainval\n33.2\n51.9 36.3 13.6 37.2 47.8 29.9 46.0 48.3 23.4 56.0 66.4\nFPN (ResNet101) [66]\ntrainval35k\n36.2\n59.1 39.0 18.2 39.0 48.2\n-\n-\n-\n-\n-\n-\n-\nMask (ResNet101+FPN) [67]\ntrainval35k\n38.2\n60.3 41.7 20.1 41.1 50.2\n-\n-\n-\n-\n-\n-\n-\nMask (ResNeXt101+FPN) [67] trainval35k\n39.8\n62.3 43.4 22.1 43.2 51.2\n-\n-\n-\n-\n-\n-\n-\nDSSD513 (ResNet101) [73]\ntrainval35k\n33.2\n53.3 35.2 13.0 35.4 51.1 28.9 43.5 46.2 21.8 49.1 66.4\nDSOD300 [74]\ntrainval\n29.3\n47.3 30.6 9.4 31.5 47.0 27.3 40.7 43.0 16.7 47.1 65.0\n* FRCN*: Fast R-CNN with multi-scale training, R-FCN*: R-FCN with multi-scale training, R-FCN**: R-FCN\nwith multi-scale training and testing, Mask: Mask R-CNN.\nX GPU. Except for ‘SS’ which is processed with CPU, the\nother procedures related to CNN are all evaluated on GPU.\nFrom Table V, we can draw some conclusions as follows.\n• By computing CNN features on shared feature maps \n(SPP-net), test consumption is reduced largely. Test time is\nfurther reduced with the unified multi-task learning (FRCN)\nand removal of additional region proposal generation stage\n(Faster R-CNN). It’s also helpful to compress the parameters\nof FC layers with SVD [91] (PAVNET and FRCN).\nTABLE V\nCOMPARISON OF TESTING CONSUMPTION ON VOC 07 TEST SET.\nMethods\nTrained on\nmAP(%)\nTest time(sec/img)\nRate(FPS)\nSS+R-CNN [15]\n07\n66.0\n32.84\n0.03\nSS+SPP-net [64]\n07\n63.1\n2.3\n0.44\nSS+FRCN [16]\n07+12\n66.9\n1.72\n0.6\nSDP+CRC [33]\n07\n68.9\n0.47\n2.1\nSS+HyperNet* [101]\n07+12\n76.3\n0.20\n5\nMR-CNN&S-CNN [110]\n07+12\n78.2\n30\n0.03\nION [95]\n07+12+S\n79.2\n1.92\n0.5\nFaster R-CNN(VGG16) [18]\n07+12\n73.2\n0.11\n9.1\nFaster R-CNN(ResNet101) [18]\n07+12\n83.8\n2.24\n0.4\nYOLO [17]\n07+12\n63.4\n0.02\n45\nSSD300 [71]\n07+12\n74.3\n0.02\n46\nSSD512 [71]\n07+12\n76.8\n0.05\n19\nR-FCN(ResNet101) [65]\n07+12+coco\n83.6\n0.17\n5.9\nYOLOv2(544*544)

[72]\n07+12\n78.6\n0.03\n40\nDSSD321(ResNet101)
[73]\n07+12\n78.6\n0.07\n13.6\nDSOD300 [74]\n07+12+coco\n81.7\n0.06\n17.4\nPVANET+
[116]\n07+12+coco\n83.8\n0.05\n21.7\nPVANET+(compress) [116]\n07+12+coco
\n82.9\n0.03\n31.3\n* SS: Selective Search [15], SS*: 'fast mode' Selective Search [16],
HyperNet*: the speed up version of\nHyperNet and PAVNET+ (compresss): PAVNET with
additional bounding box voting and compressed fully\nconvolutional layers.\n• It takes additional
test time to extract multi-scale fea-\ntures and contextual information (ION and MR-RCNN&S-
\nRCNN).\n• It takes more time to train a more complex and deeper\nnetwork (ResNet101
against VGG16) and this time con-\nsumption can be reduced by adding as many layers into
\nshared fully convolutional layers as possible (FRCN).\n• Regression based models can usually
be processed in real-'),

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SYSTEMS FOR PUBLICATION\n12\ntime at the cost of a drop in accuracy compared with region
\nproposal based models. Also, region proposal based models\ncan be modified into real-time
systems with the introduction\nof other tricks [116] (PVANET), such as BN [43], residual
\nconnections [123].\nIV. SALIENT OBJECT DETECTION\nVisual saliency detection, one of the
most important and\nchallenging tasks in computer vision, aims to highlight the\nmost
dominant object regions in an image. Numerous ap-\nplications incorporate the visual saliency
to improve their\nperformance, such as image cropping [125] and segmentation\n[126], image
retrieval [57] and object detection [66].\nBroadly, there are two branches of approaches in salient
\nobject detection, namely bottom-up (BU) [127] and top-down\n(TD) [128]. Local feature
contrast plays the central role in BU\nsalient object detection, regardless of the semantic
contents of\nthe scene. To learn local feature contrast, various local and\nglobal features are
extracted from pixels, e.g. edges [129],\nspatial information [130]. However, high-level and multi-
scale\nsemantic information cannot be explored with these low-level\nfeatures. As a result, low
contrast salient maps instead of\nsalient objects are obtained. TD salient object detection is
task-\noriented and takes prior knowledge about object categories\nto guide the generation of
salient maps. Taking semantic\nsegmentation as an example, a saliency map is generated in the
\nsegmentation to assign pixels to particular object categories via\na TD approach [131]. In a
word, TD saliency can be viewed\nas a focus-of-attention mechanism, which prunes BU salient
\npoints that are unlikely to be parts of the object [132].\nA. Deep learning in Salient Object
Detection\nDue to the significance for providing high-level and multi-\nscale feature
representation and the successful applications\nin many correlated computer vision tasks, such
as semantic\nsegmentation [131], edge detection [133] and generic object\ndetection [16], it is
feasible and necessary to extend CNN to\nsalient object detection.\nThe early work by Eleonora
Vig et al. [28] follows a\ncompletely automatic data-driven approach to perform a large-\nscale
search for optimal features, namely an ensemble of deep\nnetworks with different layers and
parameters. To address the\nproblem of limited training data, Kummerer et al. proposed the
\nDeep Gaze [134] by transferring from the AlexNet to generate\na high dimensional feature
space and create a saliency map. A\nsimilar architecture was proposed by Huang et al. to
integrate\nsaliency prediction into pre-trained object recognition DNNs\n[135]. The transfer is
accomplished by fine-tuning DNNs'\nweights with an objective function based on the saliency
\nevaluation metrics, such as Similarity, KL-Divergence and\nNormalized Scanpath Saliency.
\nSome works combined local and global visual clues to\nimprove salient object detection
performance. Wang et al.\ntrained two independent deep CNNs (DNN-L and DNN-G)\nto capture
local information and global contrast and predicted\nsaliency maps by integrating both local
estimation and global\nsearch [136]. Cholakkal et al. proposed a weakly supervised\nsaliency
detection framework to combine visual saliency from\nbottom-up and top-down saliency maps,
and refined the results\nwith a multi-scale superpixel-averaging [137]. Zhao et al.\nproposed a
multi-context deep learning framework, which\nutilizes a unified learning framework to model

global and\nlocal context jointly with the aid of superpixel segmentation\n[138]. To predict saliency in videos, Bak et al. fused two\nstatic saliency models, namely spatial stream net and tem-\nporal stream net, into a two-stream framework with a novel\nempirically grounded data augmentation technique [139].\nComplementary information from semantic segmentation\nand context modeling is beneficial. To learn internal represen-\ntations of saliency efficiently, He et al. proposed a novel su-\nperpixelwise CNN approach called SuperCNN [140], in which\nsalient object detection is formulated as a binary labeling\nproblem. Based on a fully convolutional neural network, Li\net al. proposed a multi-task deep saliency model, in which\nintrinsic correlations between saliency detection and semantic\nsegmentation are set up [141]. However, due to the conv layers\nwith large receptive fields and pooling layers, blurry object\nboundaries and coarse saliency maps are produced. Tang et\nal. proposed a novel saliency detection framework (CRPSD)\n[142], which combines region-level saliency estimation and\npixel-level saliency prediction together with three closely\nrelated CNNs. Li et al. proposed a deep contrast network\nto combine segment-wise spatial pooling and pixel-level fully\nconvolutional streams [143].\nThe proper integration of multi-scale feature maps is also\nof significance for improving detection performance. Based\non Fast R-CNN, Wang et al. proposed the RegionNet by \nperforming salient object detection with end-to-end edge pre-\nserving and multi-scale contextual modelling [144]. Liu et al.\n[27] proposed a multi-resolution convolutional neural network\n(Mr-CNN) to predict eye fixations, which is achieved by\nlearning both bottom-up visual saliency and top-down visual\nfactors from raw image data simultaneously. Cornia et al. \nproposed an architecture which combines features extracted at\ndifferent levels of the CNN [145]. Li et al. proposed a multi-\nscale deep CNN framework to extract three scales of deep \ncontrast features [146], namely the mean-subtracted region,\nthe bounding box of its immediate neighboring regions and\nthe masked entire image, from each candidate region.\nIt is efficient and accurate to train a direct pixel-wise\nCNN architecture to predict salient objects with the aids of\nRNNs and deconvolution networks. Pan et al. formulated\nsaliency prediction as a minimization optimization on the\nEuclidean distance between the predicted saliency map and\nthe ground truth and proposed two kinds of architectures\n[147]: a shallow one trained from scratch and a deeper one\nadapted from deconvoluted VGG network. As convolutional-\ndeconvolution networks are not expert in recognizing objects\nof multiple scales, Kuen et al. proposed a recurrent attentional\nconvolutional-deconvolution network (RACDNN) with several \nspatial transformer and recurrent network units to conquer\nthis problem [148]. To fuse local, global and contextual\ninformation of salient objects, Tang et al. developed a deeply-\nsupervised recurrent convolutional neural network (DSRCNN)\nto perform a full image-to-image saliency detection [149].'),

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\nX\nj=1\n\x0c\x0c\x0c ˆS(i, j) = ˆZ(i, j)\n\x0c\x0c\x0c\n(8)\nwhere ˆZ and ˆS represent the ground truth and the continuous\nsaliency map, respectively. W and H are the width and\nheight of the salient area, respectively. This score stresses\nthe importance of successfully detected salient objects over\ndetected non-salient pixels [159].\nThe following approaches are evaluated: CHM [150], RC\n[151], DRFI [152], MC [138], MDF [146], LEGS [136], DSR\n[149], MTDNN [141], CRPSD [142], DCL [143], ELD [153],\nNLDF [154] and DSSC [155]. Among these methods, CHM,\nRC and DRFI are classical ones with the best performance\n[159], while the other methods are all associated with CNN.\nF-measure and MAE scores are shown in Table VI. \nFrom Table VI, we can find that CNN based methods\nperform better than classic methods. MC and MDF combine\nthe information from local and global context to reach a\nmore accurate saliency. ELD refers to low-level handcrafted\nfeatures for complementary information. LEGS adopts generic\nregion proposals to provide initial salient regions, which may\nbe insufficient for salient detection. DSR and MT act in\ndifferent ways by introducing recurrent network and semantic\nsegmentation, which provide insights for future improvements.\nCPRSD, DCL, NLDF and DSSC are all based on multi-scale\nrepresentations and superpixel segmentation, which provide\nrobust salient regions and smooth boundaries. DCL, NLDF\nand DSSC perform the best on these four datasets. DSSC\nearns the best performance by modelling scale-to-scale short-\nconnections.\nOverall, as CNN mainly provides salient information in\nlocal regions, most of CNN based methods need to model\nvisual saliency along region boundaries with the aid of su-\nperpixel segmentation. Meanwhile, the extraction of multi-\nscale deep CNN features is of significance for measuring local\nconspicuity. Finally, it’s necessary to strengthen local con-\nnections between different CNN layers and as well to utilize\ncomplementary information from local and global context.\nV. FACE DETECTION\nFace detection is essential to many face applications and acts\nas an important pre-processing procedure to face recognition\n[160]–[162], face synthesis [163], [164] and facial expression\nanalysis [165]. Different from generic object detection, this\ntask is to recognize and locate face regions covering a very\nlarge range of scales (30-300 pts vs. 10-1000 pts). At the same\ntime, faces have their unique object structural configurations\n(e.g. the distribution of different face parts) and characteristics\n(e.g. skin color). All these differences lead to special attention\nto this task. However, large visual variations of faces, such as\nocclusions, pose variations and illumination changes, impose \ngreat challenges for this task in real applications.\nThe most famous face detector proposed by Viola and\nJones [166] trains cascaded classifiers with Haar-Like features\nand AdaBoost, achieving good performance with real-time\nefficiency. However, this detector may degrade significantly\nin real-world applications due to larger visual variations of\nhuman faces. Different from this cascade structure, Felzen-\nszwalb et al. proposed a deformable part model (DPM) for face\ndetection [24]. However, for these traditional face detection\nmethods, high computational expenses and large quantities\nof annotations are required to achieve a reasonable result.\nBesides, their performance is greatly restricted by manually\ndesigned features and shallow architecture.\nA. Deep learning in Face Detection\nRecently, some CNN based face detection approaches have\nbeen proposed [167]–[169].As less accurate localization re-\nsults from independent regressions of object coordinates, Yu\net al. [167] proposed a novel IoU loss function for predicting\nthe four bounds of box jointly. Farfade et al. [168] proposed a\nDeep Dense Face Detector (DDFD) to conduct multi-view face\ndetection, which is able to detect faces in a wide range of ori-\nentations without requirement of pose/ landmark annotations.\nYang et al. proposed a novel deep learning based face detection \nframework [169], which collects the responses from local fa-\ncial parts (e.g. eyes, nose and mouths) to address face detection\nunder severe occlusions and unconstrained pose variations. \nYang et al. [170] proposed a scale-friendly detection network\nnamed ScaleFace, which splits a large range of target scales\ninto smaller sub-ranges. Different specialized sub-networks are \nconstructed on these sub-scales and combined into a single\none to conduct end-to-end optimization. Hao et al. designed an\nefficient CNN to predict the scale distribution histogram of the\nfaces and took this histogram to guide the zoom-in and zoom-\nout of the image [171]. Since the faces are approximately\nin uniform scale after zoom, compared with other state-of-\nthe-art baselines, better performance is achieved with less\ncomputation cost. Besides, some generic detection frameworks'),

Document(metadata={'producer': 'pdfTeX-1.40.17', 'creator': 'LaTeX with hyperref package', 'creationdate': '2019-04-17T00:45:22+00:00', 'source': '..\\data\\pdf\\objectdetection.pdf', 'file_path': '..\\data\\pdf\\objectdetection.pdf', 'total_pages': 21, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2019-04-17T00:45:22+00:00', 'trapped': '', 'modDate': 'D:20190417004522Z', 'creationDate': 'D:20190417004522Z', 'page': 13}, page_content='THIS PAPER HAS BEEN ACCEPTED BY IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS FOR PUBLICATION\n14\nTABLE VI\nCOMPARISON BETWEEN STATE OF THE ART METHODS.\nDataset\nMetrics\nCHM [150]\nRC [151]\nDRFI [152]\nMC [138]\nMDF [146]\nLEGS [136]\nDSR [149]\nMTDNN [141]\nCRPSD [142]\nDCL [143]\nELD [153]\nNLDF [154]\nDSSC [155]\nPASCAL-S\nwFβ \n0.631\n0.640\n0.679\n0.721\n0.764\n0.756\n0.697\n0.818\n0.776\n0.822\n0.767\n0.831\n0.830\nMAE \n0.222\n0.225\n0.221\n0.147\n0.145\n0.157\n0.128\n0.170\n0.063\n0.108\n0.121\n0.099\n0.080\nECSSD\nwFβ \n0.722\n0.741\n0.787\n0.822\n0.833\n0.827\n0.872\n0.810\n0.849\n0.898\n0.865\n0.905\n0.915\nMAE \n0.195\n0.187\n0.166\n0.107\n0.108\n0.118\n0.037\n0.160\n0.046\n0.071\n0.098\n0.063\n0.052\nHKU-IS\nwFβ\n0.728\n0.726\n0.783\n0.781\n0.860\n0.770\n0.833\n- \n0.821\n0.907\n0.844\n0.902\n0.913\nMAE \n0.158\n0.165\n0.143\n0.098\n0.129\n0.118\n0.040\n- \n0.043\n0.048\n0.071\n0.048\n0.039\nSOD\nwFβ \n0.655\n0.657\n0.712\n0.708\n0.785\n0.707\n-\n0.781\n- \n0.832\n0.760\n0.810\n0.842\nMAE\n0.249\n0.242\n0.215\n0.184\n0.155\n0.205\n- \n0.150\n-\n0.126\n0.154\n0.143\n0.118\n* The bigger wFβ is or the smaller MAE is, the better the performance is.\nare extended to face detection with different modifications, e.g.\nFaster R-CNN [29], [172], [173].\nSome authors trained CNNs with other complementary\ntasks, such as 3D modelling and face landmarks, in a multi-\ntask learning manner. Huang et al. proposed a unified end-\nto-end FCN framework called DenseBox to jointly conduct\nface detection and landmark localization [174]. Li et al.\n[175] proposed a multi-task discriminative learning framework\nwhich integrates a ConvNet with a fixed 3D mean face model\nin an end-to-end manner. In the framework, two issues are\naddressed to transfer from generic object detection to face\ndetection, namely eliminating predefined anchor boxes by a\n3D mean face model and replacing RoI pooling layer with\na configuration pooling layer. Zhang et al. [176] proposed a \ndeep cascaded multi-task framework named MTCNN which\nexploits the inherent correlations between face detection and\nalignment in unconstrained environment to boost up detection \nperformance in a coarse-to-fine manner.\nReducing computational expenses is of necessity in real ap-\nplications. To achieve real-time detection on mobile platform,\nKalinovskii and Spitsyn proposed a new solution of frontal\nface detection based on compact CNN cascades [177]. This \nmethod takes a cascade of three simple CNNs to generate,\nclassify and refine candidate object positions progressively.\nTo reduce the effects of large pose variations, Chen et al. \nproposed a cascaded CNN denoted by Supervised Transformer\nNetwork [31]. This network takes a multi-task RPN to predict\ncandidate face regions along with associated facial landmarks\nsimultaneously, and adopts a generic R-CNN to verify the\nexistence of valid faces. Yang et al. proposed a three-stage\ncascade structure based on FCNs [8], while in each stage, a \nmulti-scale FCN is utilized to refine the positions of possible\nfaces. Qin et al. proposed a unified framework which achieves\nbetter results with the complementary information from dif-\nferent jointly trained CNNs [178].\nB. Experimental Evaluation\nThe FDDB [179] dataset has a total of 2,845 pictures in\nwhich 5,171 faces are annotated with elliptical shape. Two\ntypes of evaluations are used: the discrete score and continuous\nscore. By varying the threshold of the decision rule, the ROC\ncurve for the discrete scores can reflect the dependence of\nthe detected face fractions on the number of false alarms.\nCompared with annotations, any detection with an IoU ratio\nexceeding 0.5 is treated as positive. Each annotation is only \nassociated with one detection. The ROC curve for the contin-\nuous scores is the reflection of face localization quality.\nThe evaluated models cover DDFD [168], CascadeCNN\n[180], ACF-multiscale [181], Pico [182], HeadHunter [183],\n 0\n 0.1\n 0.2\n 0.3\n 0.4\n 0.5\n 0.6\n 0.7\n 0.8\n 0.9\n 1\n 0\n 500\n 1000\n 1500\n 2000\nTrue positive rate\nFalse positive\nDDFD

\nCascadeCNN\nACF-multiscale\nPico\nHeadHunter\nJoint Cascade\nSURF-multiview\nViola-Jones\nNPDFace\nFaceness\nCCF\nMTCNN\nConv3D\nHyperface\nUnitBox\nLDCF+\nDeepIR \nHR-ER\nFace-R-CNN\nScaleFace\n(a) Discrete ROC curves\n 0\n 0.1\n 0.2\n 0.3\n 0.4\n 0.5\n 0.6\n 0.7\n 0.8\n 0.9\n 1\n 0\n 500\n 1000\n 1500\n 2000\nTrue positive rate\nFalse positive \nDDFD\nCascadeCNN\nACF-multiscale\nPico\nHeadHunter\nJoint Cascade\nSURF-multiview \nViola-Jones\nNPDFace\nFaceness\nCCF\nMTCNN\nConv3D\nHyperface\nUnitBox\nLDCF+ \nDeepIR\nHR-ER\nFace-R-CNN\nScaleFace\n(b) Continuous ROC curves\nFig. 11. The ROC curves of state-of-the-art methods on FDDB.\nJoint Cascade [30], SURF-multiview [184], Viola-Jones [166],\nNPDFace [185], Faceness [169], CCF [186], MTCNN [176],\nConv3D [175], Hyperface [187], UnitBox [167], LDCF+ [S2],\nDeepIR [173], HR-ER [188], Face-R-CNN [172] and Scale-\nFace [170]. ACF-multiscale, Pico, HeadHunter, Joint Cascade,\nSURF-multiview, Viola-Jones, NPDFace and LDCF+ are built\non classic hand-crafted features while the rest methods are\nbased on deep CNN features. The ROC curves are shown in\nFigure 11.\nFrom Figure 11(a), in spite of relatively competitive results\nproduced by LDCF+, it can be observed that most of classic\nmethods perform with similar results and are outperformed\nby CNN based methods by a significant margin. From Figure\n11(b), it can be observed that most of CNN based methods\nearn similar true positive rates between 60% and 70% while\nDeepIR and HR-ER perform much better than them. Among\nclassic methods, Joint Cascade is still competitive. As earlier\nworks, DDFD and CCF directly make use of generated feature\nmaps and obtain relatively poor results. CascadeCNN builds\ncascaded CNNs to locate face regions, which is efficient but in-\naccurate. Faceness combines the decisions from different part\ndetectors, resulting in precise face localizations while being\ntime-consuming. The outstanding performance of MTCNN,\nConv3D and Hyperface proves the effectiveness of multi-task \nlearning. HR-ER and ScaleFace adaptively detect faces of\ndifferent scales, and make a balance between accuracy and\nefficiency. DeepIR and Face-R-CNN are two extensions of the'),

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Zhang et al. attempted to adapt generic Faster R-CNN\n[18] to pedestrian detection [203]. They modified the down-\nstream classifier by adding boosted forests to shared, high-\nresolution conv feature maps and taking a RPN to handle small\ninstances and hard negative examples. To deal with complex\nocclusions existing in pedestrian images, inspired by DPM\n[24], Tian et al. proposed a deep learning framework called\nDeepParts [204], which makes decisions based an ensemble of\nextensive part detectors. DeepParts has advantages in dealing\nwith weakly labeled data, low IoU positive proposals and\npartial occlusion.\nOther researchers also tried to combine complementary in-\nformation from multiple data sources. CompACT-Deep adopts\na complexity-aware cascade to combine hand-crafted features\nand fine-tuned DCNNs [195]. Based on Faster R-CNN, Liu et\nal. proposed multi-spectral deep neural networks for pedestrian \ndetection to combine complementary information from color\nand thermal images [205]. Tian et al. [206] proposed a task-\nassistant CNN (TA-CNN) to jointly learn multiple tasks with
\nTABLE VII\nDETAILED BREAKDOWN PERFORMANCE COMPARISONS OF\nSTATE-OF-THE-ART MODELS ON CALTECH PEDESTRIAN DATASET. ALL\nNUMBERS ARE REPORTED IN L-AMR.
\nMethod\nReasonable\nAll\nFar\nMedium\nNear\nnone\npartial\nheavy\nCheckerboards+
[198]\n17.1\n68.4\n100\n58.3\n5.1\n15.6\n31.4\n78.4\nLDCF+
+[S2]\n15.2\n67.1\n100\n58.4\n5.4\n13.3\n33.3\n76.2\nSCF+AlexNet
[210]\n23.3\n70.3\n100\n62.3\n10.2\n20.0\n48.5\n74.7\nSA-FastRCNN
[211]\n9.7\n62.6\n100\n51.8\n0\n7.7\n24.8\n64.3\nMS-CNN
[105]\n10.0\n61.0\n97.2\n49.1\n2.6\n8.2\n19.2\n60.0\nDeepParts
[204]\n11.9\n64.8\n100\n56.4\n4.8\n10.6\n19.9\n60.4\nCompACT-Deep
[195]\n11.8\n64.4\n100\n53.2\n4.0\n9.6\n25.1\n65.8\nRPN+BF
[203]\n9.6\n64.7\n100\n53.9\n2.3\n7.7\n24.2\n74.2\nF-DNN+SS
[207]\n8.2\n50.3\n77.5\n33.2\n2.8\n6.7\n15.1\n53.4\nmultiple data sources and to combine pedestrian attributes\nwith semantic scene attributes together. Du et al. proposed\na deep neural network fusion architecture for fast and robust\npedestrian detection [207]. Based on the candidate bounding\nboxes generated with SSD detectors [71], multiple binary\nclassifiers are processed parallelly to conduct soft-rejection\nbased network fusion (SNF) by consulting their aggregated\ndegree of confidences.\nHowever, most of these approaches are much more sophisti-\ncated than the standard R-CNN framework. CompACT-Deep\nconsists of a variety of hand-crafted features, a small CNN\nmodel and a large VGG16 model [195]. DeepParts contains \n45 fine-tuned DCNN models, and a set of strategies, including\nbounding box shifting handling and part selection, are required\nto arrive at the reported results [204]. So the modification and \nsimplification is of significance to reduce the burden on both\nsoftware and hardware to satisfy real-time detection demand.\nTome et al. proposed a novel solution to adapt generic object\ndetection pipeline to pedestrian detection by optimizing most\nof its stages [59]. Hu et al. [208] trained an ensemble of\nboosted decision models by reusing the conv feature maps, and\na further improvement was gained with simple pixel labelling\nand additional complementary hand-crafted features. Tome\net al. [209] proposed a reduced memory region based deep\nCNN architecture, which fuses regional responses from both\nACF detectors and SVM classifiers into R-CNN. Ribeiro et\nal. addressed the problem of Human-Aware Navigation [32]\nand proposed a vision-based person tracking system guided\nby multiple camera sensors. \nB. Experimental Evaluation\nThe evaluation is conducted on the most popular Caltech \nPedestrian dataset [3]. The dataset was collected from the\nvideos of a vehicle driving through an urban environment\nand consists of 250,000 frames with about 2300 unique\npedestrians and 350,000 annotated bounding boxes (BBs).\nThree kinds of labels, namely ‘Person (clear identifications)’,\n‘Person? (unclear identifications)’ and ‘People (large group of\nindividuals)’, are assigned to different BBs. The performance\nis measured with the log-average miss rate (L-AMR) which\nis computed evenly spaced in log-space in the range 10−2 to\n1 by averaging miss rate at the rate of nine false positives\nper image (FPPI) [3]. According to the differences in the \nheight and visible part of the BBs, a total of 9 popular settings\nare adopted to evaluate different properties of these models.\nDetails of these settings are as [3].\nEvaluated methods include Checkerboards+ [198], LDCF++\n[S2], SCF+AlexNet [210], SA-FastRCNN [211], MS-CNN'),

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'file_path': '..\\data\\pdf\\objectdetection.pdf', 'total_pages': 21, 'format': 'PDF 1.5', 'title': '', 'author': '', 'subject': '', 'keywords': '', 'moddate': '2019-04-17T00:45:22+00:00', 'trapped': '', 'modDate': 'D:20190417004522Z', 'creationDate': 'D:20190417004522Z', 'page': 15}, page_content='THIS PAPER HAS BEEN ACCEPTED BY IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS FOR PUBLICATION\n16\n[105], DeepParts [204], CompACT-Deep [195], RPN+BF \n[203] and F-DNN+SS [207]. The first two methods are based\non hand-crafted features while the rest ones rely on deep CNN\nfeatures. All results are exhibited in Table VII. From this table, \nwe observe that different from other tasks, classic handcrafted\nfeatures can still earn competitive results with boosted decision\nforests [203], ACF [197] and HOG+LUV channels [S2]. As\nan early attempt to adapt CNN to pedestrian detection, the\nfeatures generated by SCF +AlexNet are not so discriminant\nand produce relatively poor results. Based on multiple CNNs, \nDeepParts and CompACT-Deep accomplish detection tasks via\ndifferent strategies, namely local part integration and cascade\nnetwork. The responses from different local part detectors \nmake DeepParts robust to partial occlusions. However, due to\ncomplexity, it is too time-consuming to achieve real-time de-\ntection. The multi-scale representation of MS-CNN improves\naccuracy of pedestrian locations. SA-FastRCNN extends Fast\nR-CNN to automatically detecting pedestrians according to\ntheir different scales, which has trouble when there are partial\nocclusions. RPN+BF combines the detectors produced by\nFaster R-CNN with boosting decision forest to accurately\nlocate different pedestrians. F-DNN+SS, which is composed\nof multiple parallel classifiers with soft rejections, performs\nthe best followed by RPN+BF, SA-FastRCNN and MS-CNN.\nIn short, CNN based methods can provide more accurate \ncandidate boxes and multi-level semantic information for\nidentifying and locating pedestrians. Meanwhile, handcrafted\nfeatures are complementary and can be combined with CNN\nto achieve better results. The improvements over existing CNN\nmethods can be obtained by carefully designing the framework\nand classifiers, extracting multi-scale and part based semantic\ninformation and searching for complementary information\nfrom other related tasks, such as segmentation.\nVII. PROMISING FUTURE DIRECTIONS AND TASKS\nIn spite of rapid development and achieved promising\nprogress of object detection, there are still many open issues\nfor future work.\nThe first one is small object detection such as occurring\nin COCO dataset and in face detection task. To improve\nlocalization accuracy on small objects under partial occlusions,\nit is necessary to modify network architectures from the\nfollowing aspects. \n• Multi-task joint optimization and multi-modal infor-\nmation fusion. Due to the correlations between different\ntasks within and outside object detection, multi-task joint\noptimization has already been studied by many researchers\n[16] [18]. However, apart from the tasks mentioned in\nSubs. III-A8, it is desirable to think over the characteristics\nof different sub-tasks of object detection (e.g. superpixel\nsemantic segmentation in salient object detection) and ex-\ntend multi-task optimization to other applications such as\ninstance segmentation [66], multi-object tracking [202] and\nmulti-person pose estimation [S4]. Besides, given a specific\napplication, the information from different modalities, such\nas text [212], thermal data [205] and images [65], can be\nfused together to achieve a more discriminant network.\n• Scale adaption. Objects usually exist in different scales,\nwhich is more apparent in face detection and pedestrian \ndetection. To increase the robustness to scale changes, it\nis demanded to train scale-invariant, multi-scale or scale-\nadaptive detectors. For scale-invariant detectors, more pow-\nerful backbone architectures (e.g. ResNext [123]), negative\nsample mining [113], reverse connection [213] and sub-\ncategory modelling [60] are all beneficial. For multi-scale\ndetectors, both the FPN [66] which produces multi-scale\nfeature maps and Generative Adversarial Network [214]\nwhich narrows representation differences between small ob-\njects and the large ones with a low-cost architecture provide\ninsights into generating meaningful feature pyramid. For\nscale-adaptive detectors, it is useful to combine knowledge\ngraph [215], attentional mechanism [216], cascade network\n[180] and scale distribution estimation [171] to detect ob-\njects adaptively.\n• Spatial correlations and contextual modelling. Spatial \ndistribution plays an important role in object detection. So\nregion proposal generation and grid regression are taken\nto obtain probable object locations. However, the corre-\nlations between multiple proposals and object categories\nare ignored. Besides, the global structure information is\nabandoned by the position-sensitive score maps in R-FCN.\nTo solve these problems, we can refer to diverse subset\nselection [217] and sequential reasoning tasks [218]

for\npossible solutions. It is also meaningful to mask salient parts\nand couple them with the global structure in a joint-learning\nmanner [219].\nThe second one is to release the burden on manual labor and\naccomplish real-time object detection, with the emergence of\nlarge-scale image and video data. The following three aspects\ncan be taken into account.\n• Cascade network. In a cascade network, a cascade of\ndetectors are built in different stages or layers [180], [220].\nAnd easily distinguishable examples are rejected at shallow\nlayers so that features and classifiers at latter stages can\nhandle more difficult samples with the aid of the decisions\nfrom previous stages. However, current cascades are built in\na greedy manner, where previous stages in cascade are fixed\nwhen training a new stage. So the optimizations of different\nCNNs are isolated, which stresses the necessity of end-to-\nend optimization for CNN cascade. At the same time, it\nis also a matter of concern to build contextual associated \ncascade networks with existing layers.\n• Unsupervised and weakly supervised learning. It's \nvery time consuming to manually draw large quantities\nof bounding boxes. To release this burden, semantic prior\n[55], unsupervised object discovery [221], multiple instance\nlearning [222] and deep neural network prediction [47] can\nbe integrated to make best use of image-level supervision to\nassign object category tags to corresponding object regions\nand refine object boundaries. Furthermore, weakly annota-\ntions (e.g. center-click annotations [223]) are also helpful\nfor achieving high-quality detectors with modest annotation\nefforts, especially aided by the mobile platform.\n• Network optimization. Given specific applications and\nplatforms, it is significant to make a balance among speed,'),

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also meaningful for the\ndevelopments in neural networks and related learning systems,\nwhich provides valuable insights and guidelines for future\nprogress.\nACKNOWLEDGMENTS\nThis research was supported by the National Natural Sci-\nence Foundation of China (No.61672203 & 61375047 &\n91746209), the National Key Research and Development Pro-\ngram of China (2016YFB1000901), and Anhui Natural Sci-\nence Funds for Distinguished Young Scholar (No.170808J08).\nREFERENCES\n[1] P. F. Felzenszwalb, R. B. Girshick, D. Mcallester, and D. Ramanan,\n“Object detection with discriminatively trained part-based models,”\nIEEE Trans. Pattern Anal. Mach. Intell., vol. 32, no. 9, p. 1627, 2010.\n[2] K. K. Sung and T. Poggio, “Example-based learning for view-based\nhuman face detection,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 20,\nno. 1, pp. 39–51, 2002.\n[3] C. Wojek, P. Dollar, B. Schiele, and P. Perona, “Pedestrian detection:\nAn evaluation of the state of the art,” IEEE Trans. Pattern Anal. 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Pattern Recognition & Intelligent System\nat University of Science and Technology, China, in \n2007. From April 2008 to November 2009, he held a\npostdoctoral position in image processing in CNRS\nUMR6168 Lab Sciences de lInformation et des\nSyst`emes, France. From January 2013 to December\n2014, he held a research fellow position in image\nprocessing at the Department of Computer Science\nof Hongkong Baptist University, Hongkong, China.\nHis research is about pattern recognition, image processing, and computer\nvision.\nPeng Zheng is a Ph.D. candidate at Hefei Uni-\nversity of Technology since 2010. He received his\nBachelor's degree in 2010 from Hefei University of\nTechnology. His interests cover pattern recognition, \nimage processing and computer vision.\nShou-tao Xu is a Master student at Hefei University \nof Technology. His research interests cover pattern\nrecognition, image processing, deep learning and\ncomputer vision.\nXindong Wu is an Alfred and Helen Lamson En-\ndowed Professor in Computer Science, University\nof Louisiana at Lafayette (USA), and a Fellow of\nthe IEEE and the AAAS. He received his Ph.D.\ndegree in Artificial Intelligence from the University \nof Edinburgh, Britain. His research interests include\ndata mining, knowledge-based systems, and Web in-\nformation exploration. He is the Steering Committee\nChair of the IEEE International Conference on Data\nMining (ICDM), the Editor-in-Chief of Knowledge\nand Information Systems (KAIS, by Springer), and\na Series Editor of the Springer Book Series on Advanced Information and\nKnowledge Processing (AI&KP). He was the Editor-in-Chief of the IEEE\nTransactions on Knowledge and Data Engineering (TKDE, by the IEEE\nComputer Society) between 2005 and 2008.'),

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